MILITARY SPECIFICATION SHEET

MODULES, STANDARD ELECTRONIC PROGRAM

MEMORY, READ ONLY, DIGITAL

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the procurement of an electronic digital read only memory module

1 2 Module description The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Memory, read only
Family	Digital
Key code	As specified in 3 5 1
Size	1A
Weight	35 grams maximum
Failure rate	2 4 failures/10^{6} hours maximum
Environment	Class I as specified in MIL-M-28787
Maximum power dissipation	2 86 watts (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{cc} maximum
Input current, A10-A18, A20-A28	
Low level (V_{IL} = 0.0 V)	-0.88 mA maximum
(V_{IL} = 0.45 V)	-0.8 mA maximum
High level (V_{IH} = 2.4 V)	80 µA maximum
(V_{IH} = 5.5 V)	2.0 mA maximum
Input current, E1 and E2	
Low level (V_{IL} = 0.0 V)	-0.88 mA maximum
(V_{IL} = 0.45 V)	-0.8 mA maximum
High level (V_{IH} = 2.4 V)	80 µA maximum
(V_{IH} = 5.5 V)	40 mA maximum
(V_{IH} = 4.5 V)	2.0 mA maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, D C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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Input clamp diode voltage (V_{CC} = 4.5 V, I_{Clamp} = -5 mA)	-1.5 V maximum
Output voltage	
Low level (I_{OL} = 8.0 mA)	0.5 V maximum
High level (I_{OH} = -900 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	8.0 mA
High level (V_{OH} = 2.4 V)	-900 µA
Disable output current	
Low level (V_{OL} = 0.5 V)	±100 µA maximum
High level (V_{OH} = 2.4 V)	±100 µA maximum
Input capacitance, A10-A16, A20-A28	30 pF maximum
Input capacitance, E1 and E2	35 pF maximum
Output capacitance, D10-D17, D20-D27	20 pF maximum
Dynamic characteristics (see figure 1)	
Propagation delay time	
t_{AA} - Address to data out, C_L = 50 pF	95.0 ns maximum
t_{AA} - Address to data out, C_L = 150 pF	105.0 ns maximum
t_{EA} - Enable/disable, C_L = 50 pF	45.0 ns maximum
t_{EA} - Enable/disable, C_L = 150 pF	55.0 ns maximum
Enable recovery time	
Low-state to high-impedance state (t_{ERL}), C_L = < 10 pF	50 ns maximum
High-state to high-impedance state (t_{ERH}), C_L = < 10 pF	50 ns maximum

1.3 Module function. The read only memory module is a plug-in Standard Hardware Program (SHP) electronic module with 0.100 inch (2.54 mm) grid center contacts. The module is organized as two independent 512-word by 8-bit memories. These memories may be interconnected through connector wiring to form a memory of either 512 by 16 or 1,024 by 8.

1.3.1 Input to output relationship. Each read only memory module has the input to output relationship of table I. The following describes the essential module operation. The module contains two independent 512-word by 8-bit read only memories as shown on figure 4. When the enable of one of the 512-word by 8-bit memories goes to the low level, the data stored at the selected address input appears at the eight data outputs of that memory. When the enable of one of the memories is high, the eight output lines of that memory are disabled to a high-impedance state.

1.3.2 Paralleling inputs and outputs. Corresponding address inputs of the two address groups must be paralleled for nonindependent operation of the two memory sections.

The 512 by 16 organization is achieved by paralleling the corresponding addresses of the two sections and connecting both circuit enable lines together, as shown on figure 2.

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The 1024 by 8 organization is achieved by paralleling the corresponding addresses of the two sections and the corresponding outputs of D10 through D17 with D20 through D27, as shown on figure 3

A number of memory outputs can be connected to a common bus. All of the devices except one are placed in the high-impedance state, and the selected device is enabled When outputs of several modules are bussed together, all address lines should be addressed using synchronous clocking

TABLE I. Input to output relationship

E1	E2	Outputs	
		D10-D17	D20-D27
H	H	High Z	High Z
H	L	High Z	Read data at A20-A28
L	H	Read data at A10-A18	High Z
L	L	Read data at A10-A18	Read data at A20-A28

NOTES 1 H = High-level input
2 L = Low-level input
3 High Z = High-impedance state

2 APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787 - Modules Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules

MIL-STD-1388 - Test Equipment for the Standard Electronic Modules Program

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue Philadelphia, PA 19120

PUBLICATIONS

MILITARY

3202946G1 - SEM Memory Module Test Adapter

(All Ordnance Data publications shall be ordered from the Commanding Officer, Technical Documents Department, Naval Ordnance Station, Louisville, KY 40214)

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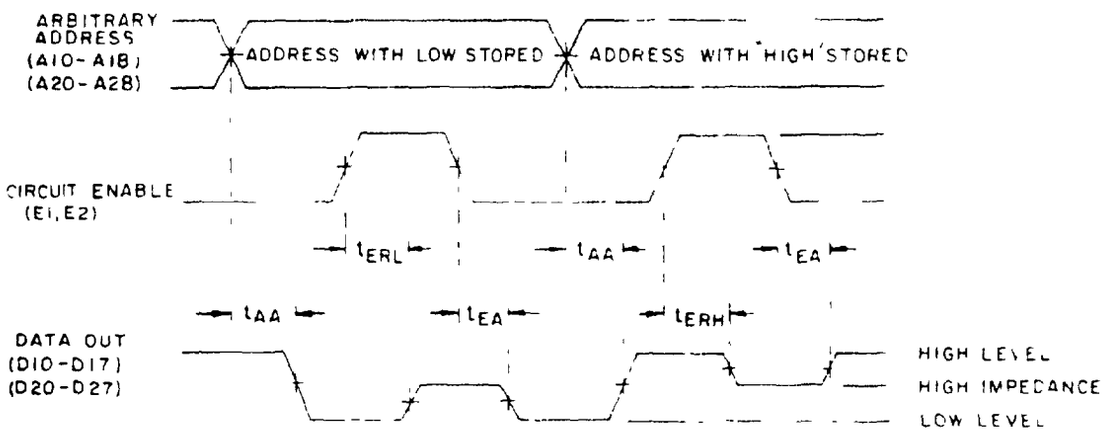


FIGURE 1 Dynamic characteristics waveforms

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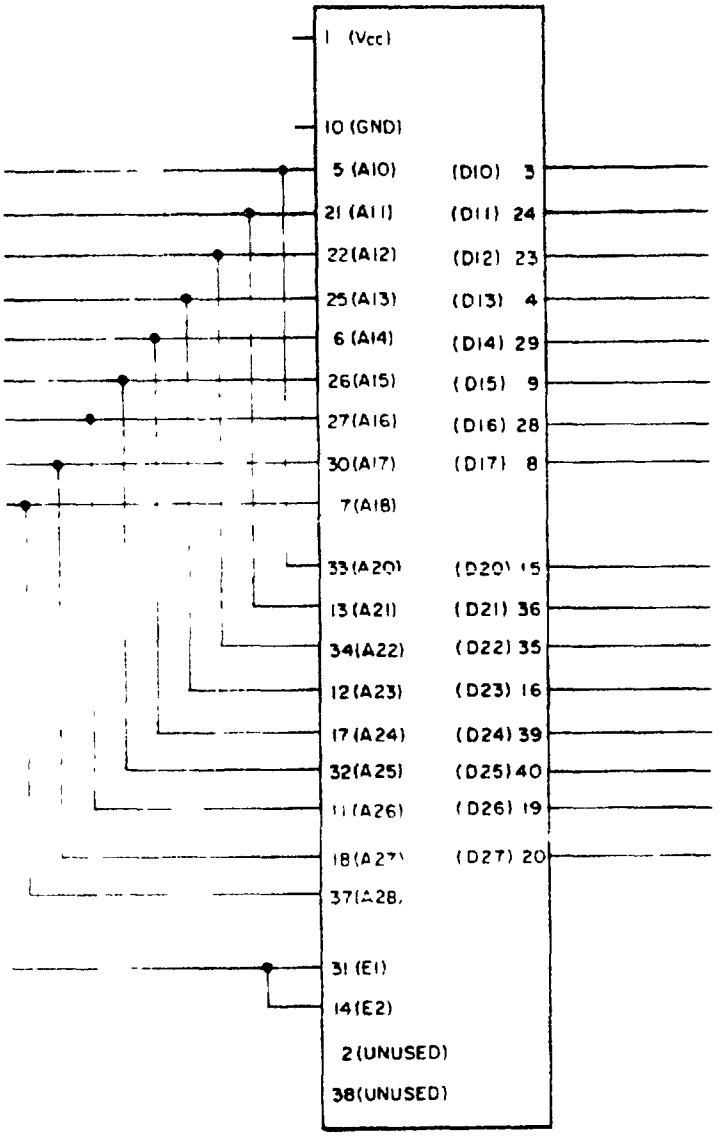


FIGURE 2. Pin interconnection for 512-word by 16-bit configuration

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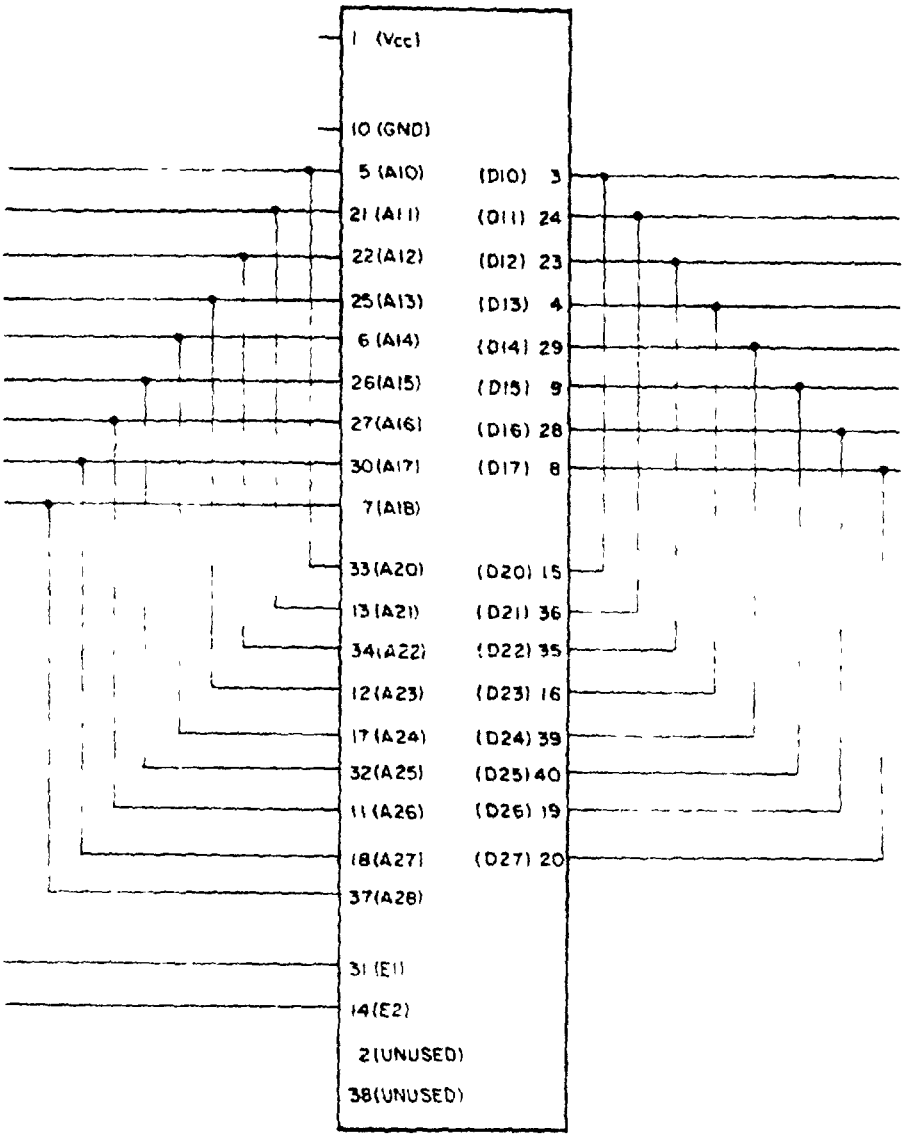
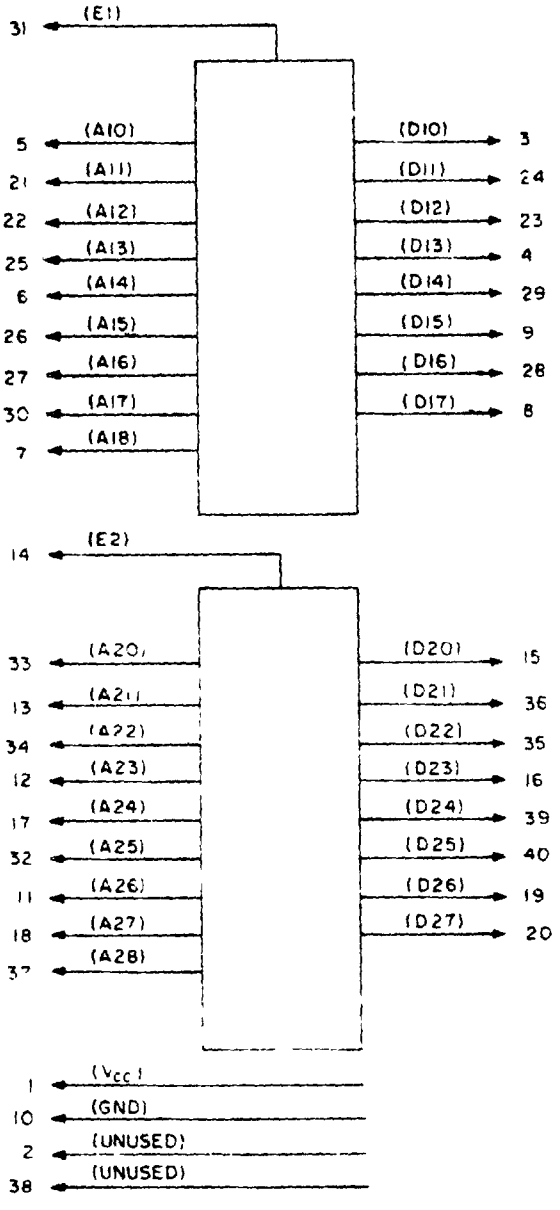


FIGURE 3 Pin interconnection for 1024-word by 8-bit configuration

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NOTES
1. A10-A18, A20-A28 = Address inputs
2. D10-D17, D20-D27 = Data outputs
3. E1, E2 = Circuit enable inputs
4. V_{CC} = 5.0 V ±10%
5. GND = 0 V REF

FIGURE 4 Functional diagram, two by 512-word by 8-bit read only memory module

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3 REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.1.1 Application requirements. This specification establishes the general requirements and test procedures for a read only memory (ROM) module. When a SEMP user determines a requirement for an ROM, the user shall prepare a design approval request (DAR) and submit the DAR to the SEMP Design Review Activity (SEMP-DRA) in accordance with MIL-STD-1389. The module user shall prepare a separate module specification. The module specification shall specify, as a minimum, the module key code as assigned by the SEMP-DRA, the firmware pattern (truth table) detailed module marking, and requirements specified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 2.4 failures per million operating hours when predicted in accordance with MIL-M-28787.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key Code. As specified in the user's detailed specification.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/74 except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The logic diagram and pin assignment shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

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TABLE II Module electrical requirements

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation	III	S	10 0	∞	1.0	∞	10 0	∞	1 0	∞	MΩ
A2, A3	Unused pin isolation	III	X	10 0	∞	1.0	∞	10 0	∞	1.0	∞	MΩ
B1-B16	V_{out} (low-loaded)	IV	S	0.0	1 0	0 0	1 0	0 0	1 0	0 0	1.0	V
C1-C18	Input breakdown current, A10-A18, A20-A28	V	S	0.0	2.0	0 0	2.0	0.0	2 0	0 0	2 0	mA
C19-C20	Input breakdown current, F1, E2	V	S	0.0	40 0	0 0	40.0	0 0	40 0	0 0	40 0	mA
C21,C22	Input breakdown current, E1, F2 (4,5V)	V	S	0 0	2.0	0 0	2.0	0.0	2 0	0.0	2.0	mA
D1-D16	Output short circuit current	VI	X	-8.0	-90 0	-8 0	-90.0	-8.0	-90 0	-8.0	-90 0	mA
E1-E16	Output leakage current, low level	VII	X	-100	100	-100	100	-100	100	-100	100	µA
F1-F16	Output leakage current, high level	VIII	X	-100	100	-100	100	-100	100	-100	100	µA
G1-G20	Input clamp diode	IX	X	-0 2	-1.5	-0 2	-1.5	-0.2	-1 5	-0 2	-1.5	V
H1-H16	Static output voltage, low level	X	X	0.0	0.50	0.0	0.50	0 0	0.50	0 0	0 50	V
J1-J16	Static output voltage, high level	XI	X	2 4	4.5	2 4	4.5	2.4	4.5	2 4	4 5	V
K1-K20	Input current, high-level, A10-A18, A20-A28, E1, E2	XII	X	-100	80.0	-100	80.0	-100	80.0	-100	80.0	µA
L1-L20	Input current, low-level, A10-A18, A20-A28, E1, E2	XIII	X	0.0	-0.88	0.0	-0.88	0.0	-0.88	0 0	-0 88	mA
M	Power supply current	XIV	X	5.0	520.0	5.0	520.0	5.0	520 0	5 0	520 0	mA

See notes at end of table

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TABLE II Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N16	Transition time (t_{TLH}), C_L = 50 pF	XV	S	4	40	4	40	4	45	4	45	ns
P1-P16	Transition time (t_{THL}), C_L = 50 pF	XV	S	4	18	4	18	4	20	4	20	ns
N17-N32	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	4	50	4	50	4	55	4	55	ns
P17-P32	Transition time (t_{THL}), C_L = 150 pF	XV	Q	4	20	4	20	4	25	4	25	ns
Q1-Q16	Enable recovery time (t_{ERL}), D10-D17, D20-D27	XVI	S	5	50	5	50	5	50	5	50	ns
Q17-Q32	Enable recovery time (t_{ERH}), D10-D17, D20-D27	XVI	S	5	50	5	50	5	50	5	50	ns
R1, R2	Dynamic tests	XVII	X	← No fail →								
R3, R4	Dynamic tests	XVII	Q	← No fail →								
S1, S2	Input crosstalk, low level	XVIII	Q	0	400	0	400	0	400	0	400	mV
S3, S4	Input crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	mV
T	Functional test	XIX	X	← No fail →								
U	Power supply transient amplitude	XX	Q	0	400	0	400	0	400	0	400	mV

See notes at end of table

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TABLE II. Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V	Power supply filter capacitance	XXI	S	1 6	3.0	1 6	3.0	1.6	3.0	1 6	3 0	uF
W1-W18	Input capacitance, A10-A18, A20-A28	XXII	Q	2 0	30 0	2 0	30.0	2.0	30.0	2 0	30 0	pF
W19,W20	Input capacitance, E1,E2	XXII	Q	2.0	35 0	2 0	35.0	2 0	35 0	2 0	35.0	pF
X1-X16	Output capacitance	XXII	Q	2 0	20 0	2 0	20 0	2 0	20 0	2.0	20 0	pF

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance test.
2. (S) parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C
3. (Q) parameters shall be tested during qualification testing only

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3 9 Power supply voltage The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{cc}) of 5.0 ± 0.5 Vdc, except that the limits for V_{out} (low-loaded) apply only for a power supply voltage of 7.0 Vdc +0, -1 percent.

4 QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply

4 2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4.3 Test equipment The following test equipment is necessary to conduct acceptance tests for this module Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665	No required
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
DC ammeter	Item X01	1
DC voltmeter	Item V03	1
Pulse generator	Item J01	1
Power supply, DC	Item P01	4
Megohm bridge	Item R06	1
Resistance bridge	Item R02	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
LSI tester with SEM memory module test adapter - 3202946G1	Item U03	1
Resistors		
1Ω ± 1%, 1/2 W		1
51Ω ± 5%, 1/4 W		1
51Ω ± 5%, 1/2 W		2
51Ω ± 5%, 1 W		1
51 1Ω ± 1%, 1 W		1
150Ω ± 1%, 1/8 W		1
619Ω ± 1%, 1/8 W		16
619Ω ± 1%, 1/4 W		1
2 02 kΩ ± 1%, 1/8 W		1
2 2 kΩ ± 10%, 1/8 W		3
2 61 kΩ ± 1%, 1/8 W		16
Values as required for figure 19, 1/8 W		16
Capacitors		
0.1 µF ± 10%		1
6.8 µF ± 10%		1
Value as required for figure 19, note 4		16
Value as required for figure 19, note 5		16
Value as required for figure 19, note 6		16
Value as required for figure 30, note 5		2
Diodes		
IN3600, IN4150, or equivalent		16

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Components	No required
5400, or equivalent	1
9316, or equivalent	3
Switch, SPDT	9
Pushbutton, DPST	1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified

4.4 Test procedures. The module shall be tested as specified in table II

4 4 1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787

4 4 2 Order of testing Test conditions A through Y shall be performed in any order with the following exceptions If conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows A, B, C, D, where applicable, followed by the remaining tests in the sequence

4 5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4 5 1 Operating temperature test The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through X in table II shall be performed in the order specified in 4 4.2 using the initial limits as 25°C ± 5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 5

4 5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 8.

4 5.2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 9 Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column and test condition V shall be performed using the initial limits at 25° ± 5°C ambient.

4 5 3 Life tests The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 7 Following the life tests, electrical tests D through V shall be performed in the order specified in 4 4.2 using the 0° to 60°C operating temperature end-of-life requirements of table II After the module has been returned to 25°C, electrical tests D through V shall be performed in the order specified in 4 4.2 using the 25° ± 5°C end-of-life requirements of table II

5. PACKAGING

5.1 Packaging shall be in accordance with MIL-M-28787.

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6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-1331.

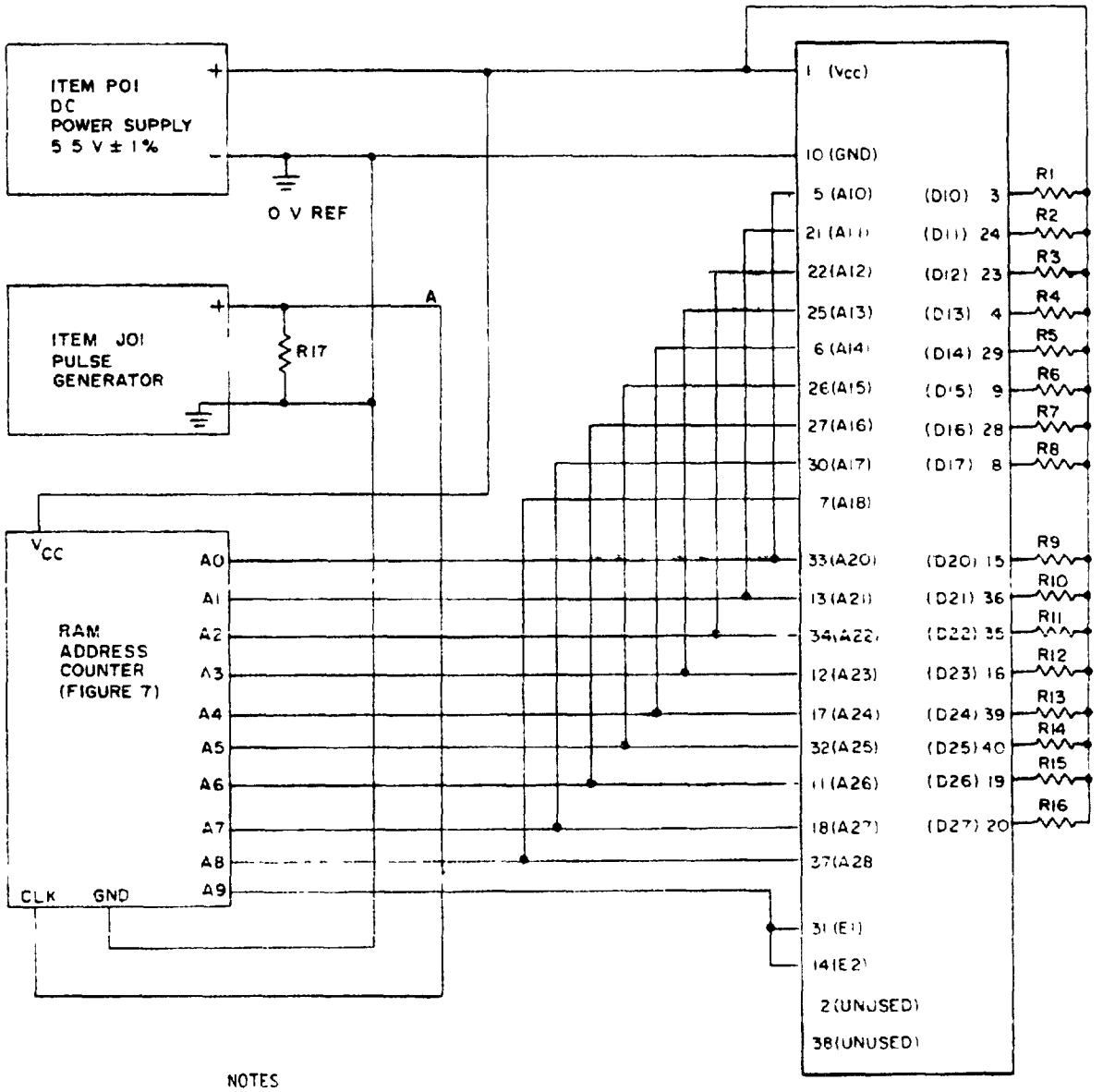
6 2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life

Preparing activity
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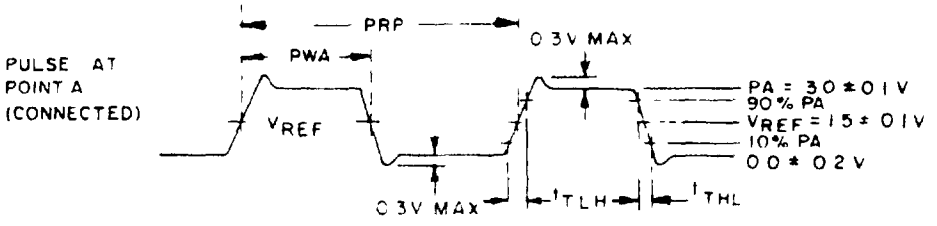
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NOTES
1 R1-R16 = 619 ±1%, 1/8 W
2 R17 = 510 ±5%, 1 W
3 See figure 6 for life-test waveform

FIGURE 5 Operating temperature and life test circuit

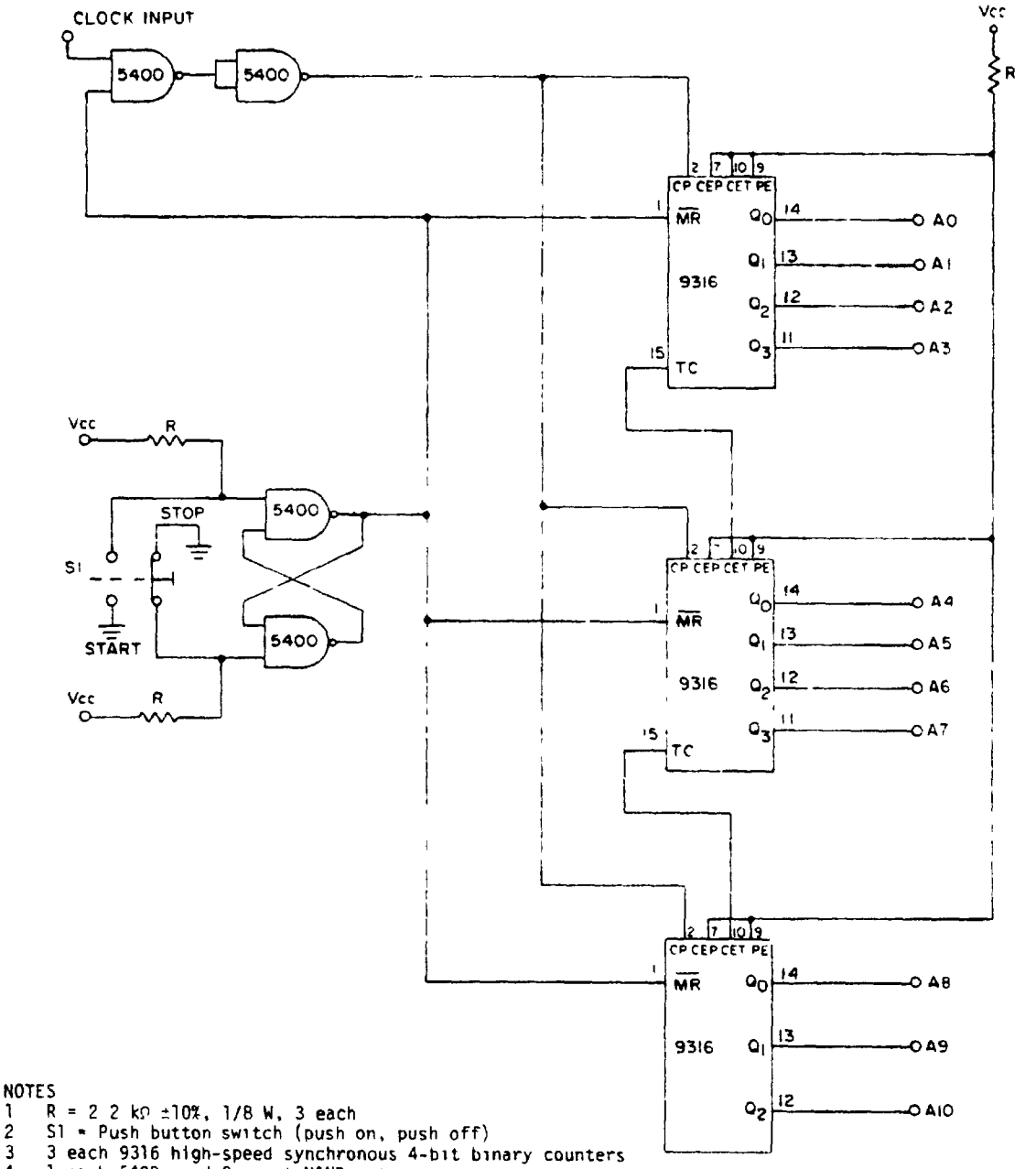
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NOTE Pulse parameters for item J01 pulse generator
t_{TLH}, t_{THL} = 20 ±10 ns
PWA = 100 ±10 ns
PRP = 200 ±10 ns

FIGURE 6 Operating temperature and life test waveform

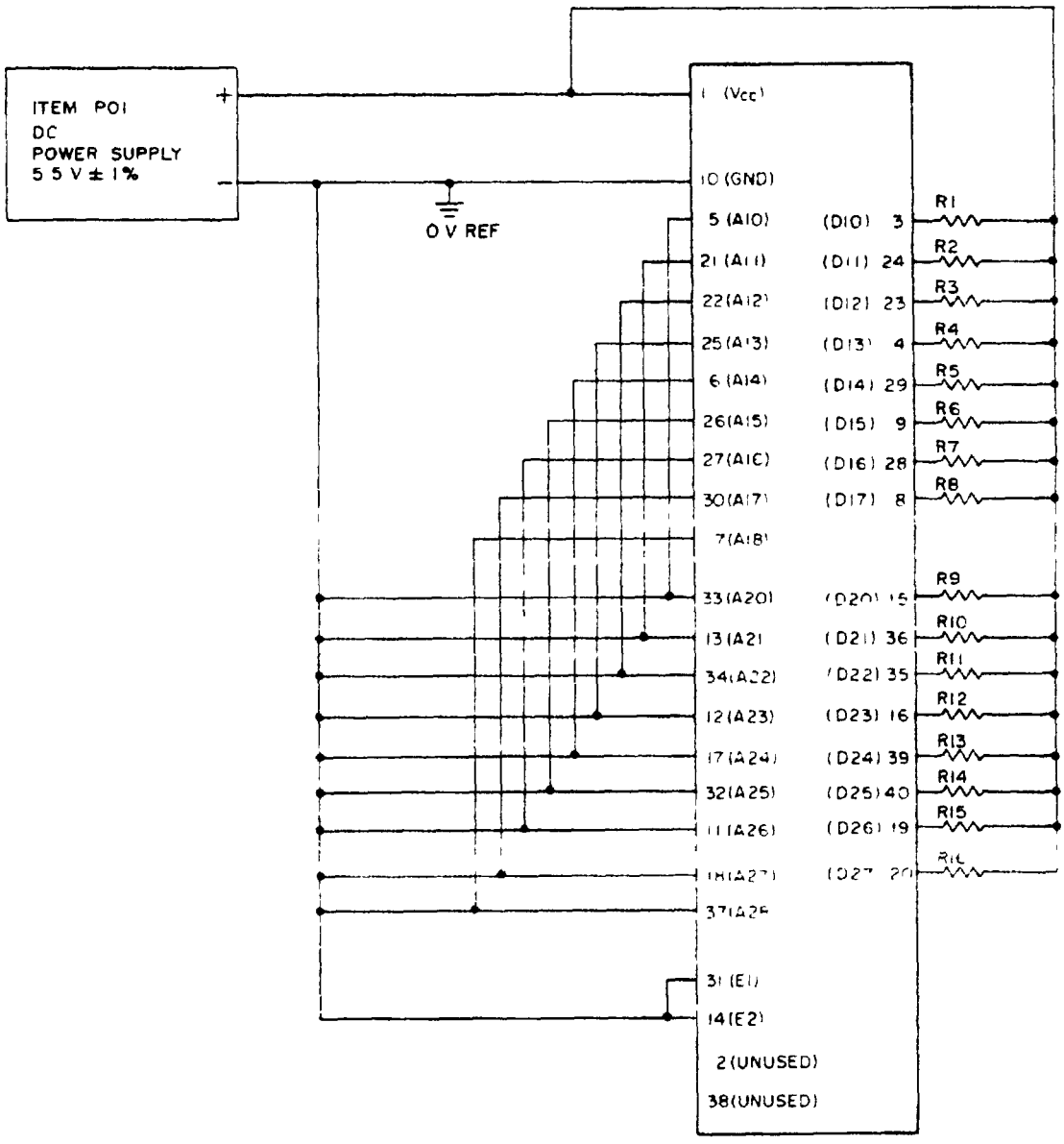
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NOTES
1 R = 2 2 kΩ ±10%, 1/8 W, 3 each
2 S1 = Push button switch (push on, push off)
3 3 each 9316 high-speed synchronous 4-bit binary counters
4 1 each 5400 quad 2-input NAND gate

FIGURE 7 RAM address counter

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NOTE R1-P16 = 619 Ω, 1/8 W

FIGURE 8 Component temperature test circuit

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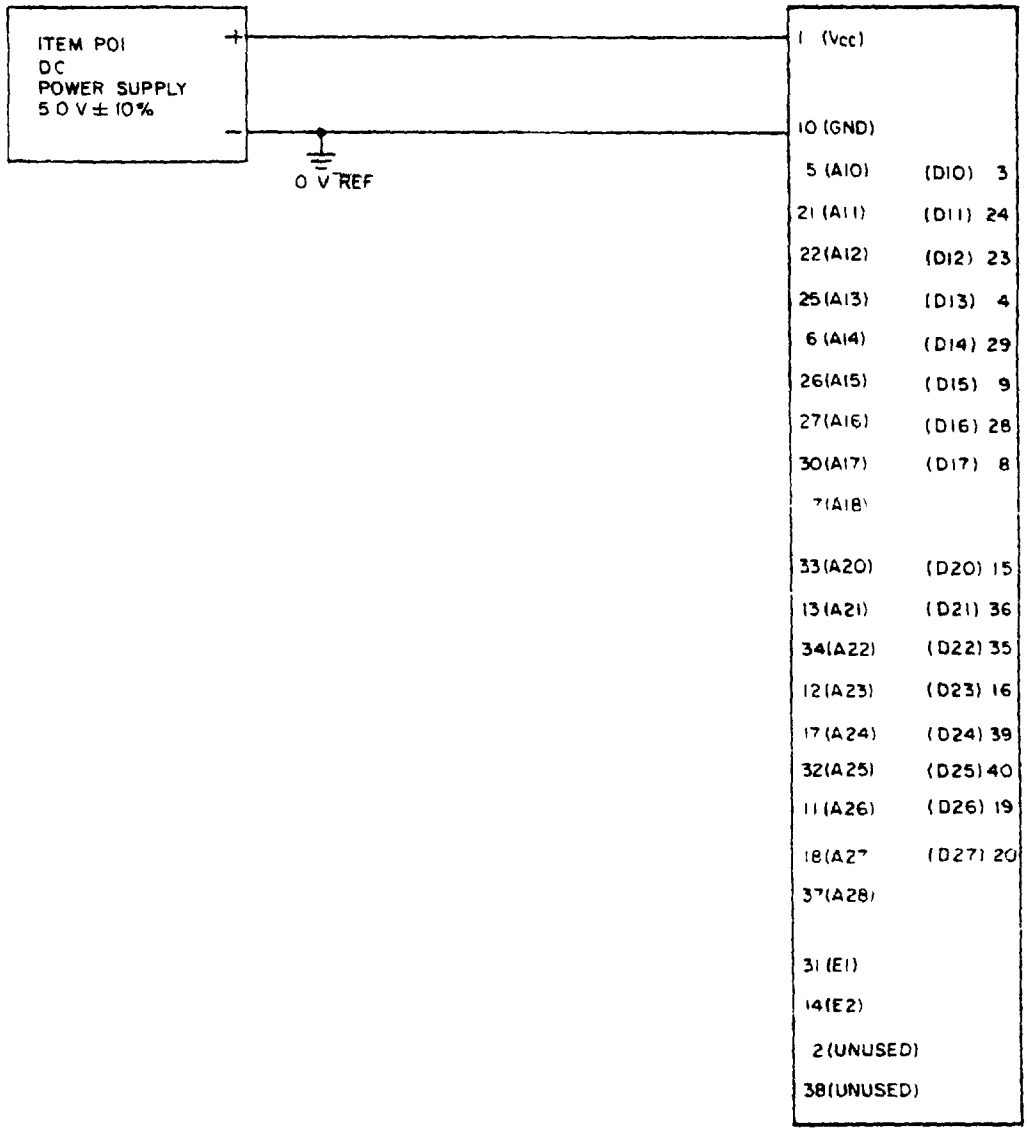


FIGURE 9 Vibration test circuit

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TABLE III Isolation test

Test condition	Test	Terminals		Limits at +25°C Ambient		Unit	Test figure
		+ to pin	- to pin	Min	Max		
A1	Isolation	A, 2, 38	Module fin (see note)	10.0	∞	MΩ	10
A2	Unused pin isolation	A, 2	38	10.0	∞	MΩ	10
A3	Unused pin isolation	A, 38	2	10.0	∞	MΩ	10

NOTE Penetrate the anodized finish on the module fin with a sharply pointed probe. Connect the negative lead of the megohm bridge to the probe.

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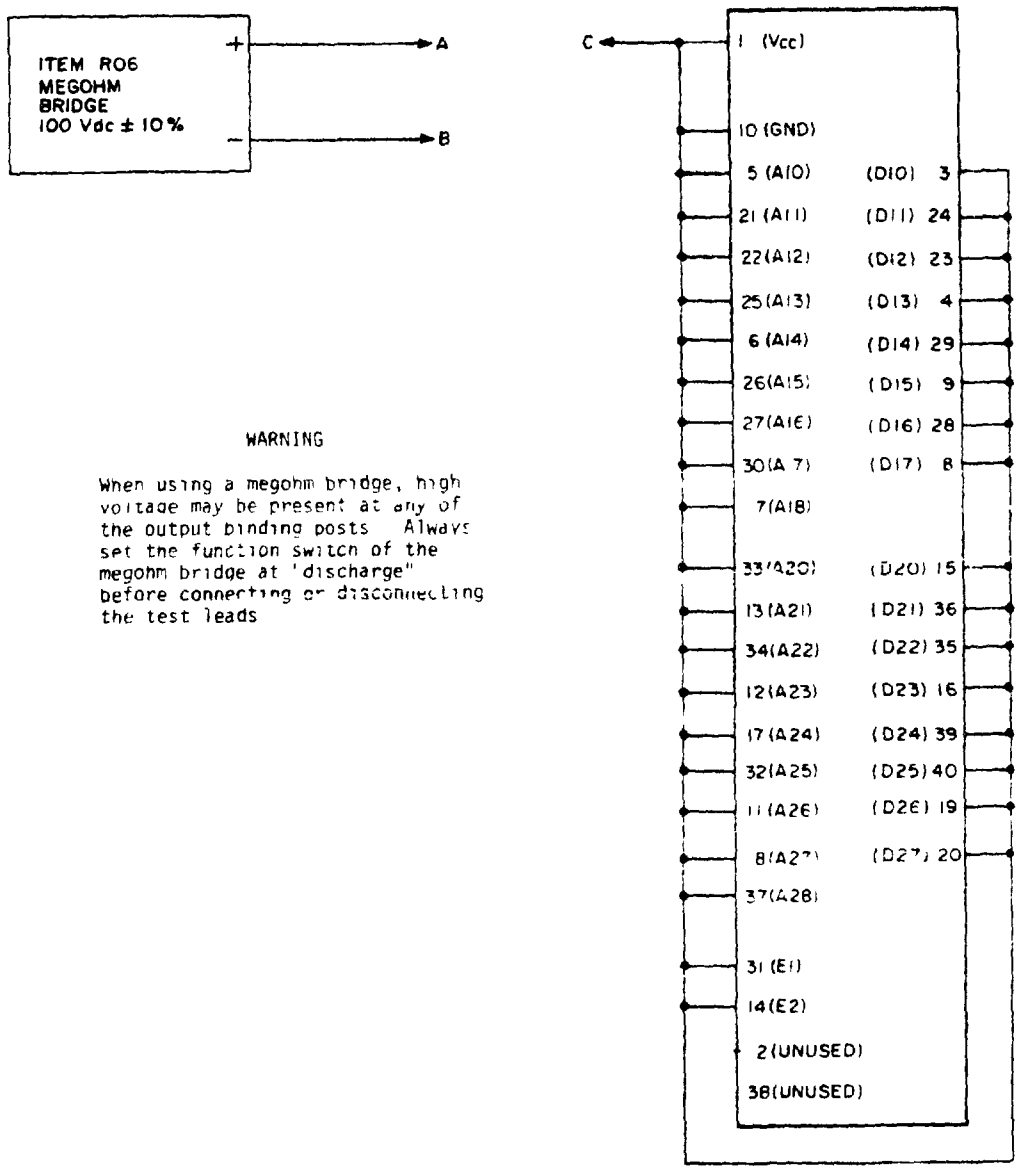


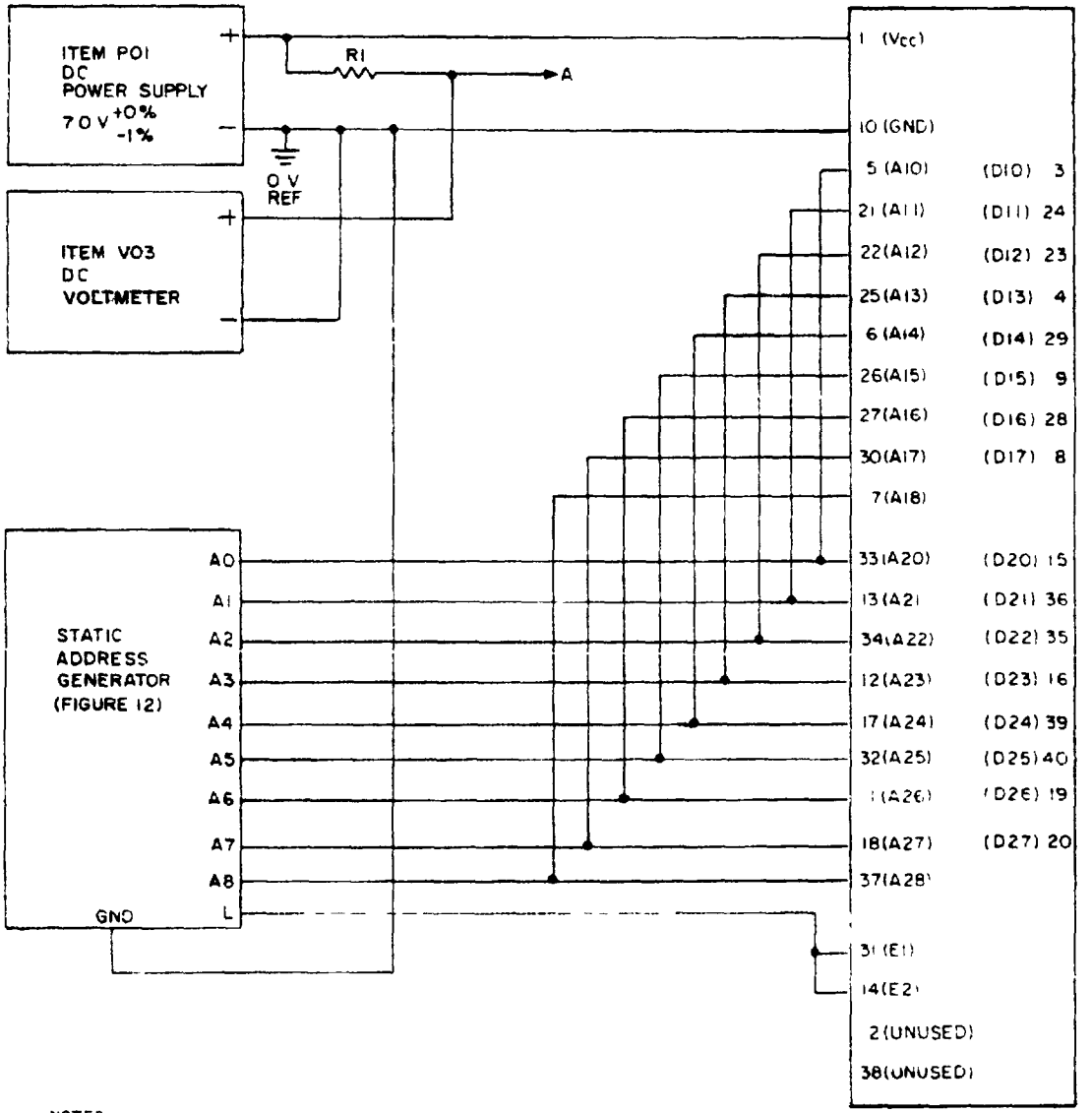
FIGURE 10 Isolation test circuit for table III

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TABLE IV V_{out} (low-loaded) test

Test condition	Parameter	Connect to pin	Initial limits 25° +5°C Ambient		Unit	Test figure
			Min	Max		
B1	D10 V_{out} (low-loaded)	3	0 0	1 0	V	11
B2	D11	24				
B3	D12	23				
B4	D13	4				
B5	D14	29				
B6	D15	9				
B7	D16	28				
B8	D17	8				
B9	D20	15				
B10	D21	36				
B11	D22	35				
B12	D23	16				
B13	D24	39				
B14	D25	40				
B15	D26	19				
B16	D27 V_{out} (low-loaded)	20	0 0	1 0	V	11

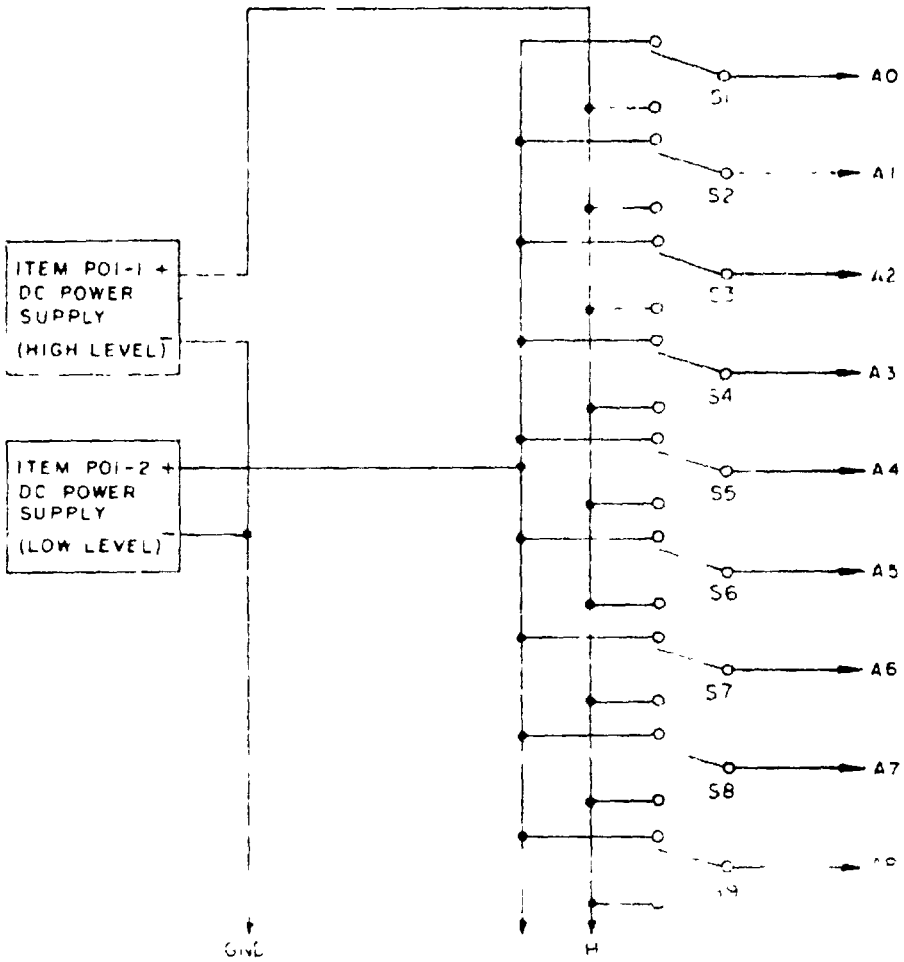
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NOTES
1. R1 = 619 ±1%, 1/4 W
2. For test condition B only, set the high level and low level of the static address generator to
   H = 3.5 ±0.5 V, L = 0.0 V
3. Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a low level before connecting point A. If an output is high for all addresses, omit this test for that output.

FIGURE 11. V_{out} (low-loaded) test circuit for table IV.

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NOTES
1 High level (H) is defined as 2.0 +0.1, -0.0 V
2 Low level (L) is defined as 0.8 +0.0, -0.1 V
3 S1-S9 = SPDT

FIGURE 12 Static address generator

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TABLE V Input breakdown current test

Test condition	Test	Input	Common		Thermal limits -55°C to +125°C Ambient		Unit	Test figure
			All pins	Pin to pin	Min	Max		
C1	A10	Input breakdown current	9,7,11,12,13, 14,17,18,21, 22,25,26,27,30, 31,32,33,34,37	5	0.0	2.0	mA	13
C2	A11		5,6,7,11,12, 13,14,17,18,22, 25,26,27,30,31, 32,33,34,37	21				
C3	A12		5,6,7,11,12, 13,14,17,18,21, 25,26,27,30,31, 32,33,34,37	22				
C4	A13		5,6,7,11,12, 13,14,17,18,21, 22,26,27,30,31, 32,33,34,37	25				
C5	A14		5,7,11,12,13, 14,17,18,21, 22,25,26,27, 30,31,32,33, 34,37	6				
C6	A15		5,6,7,11,12,13, 14,17,18,21,22, 25,27,30,31,32, 33,36,37	26				
C7	A16	Input breakdown current	5,6,7,11,12,13, 14,17,18,21,22, 25,26,30,31,32, 33,36,37	27	0.0	2.0	mA	13

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:54Z
Check the source to verify that this is the current version before use.

TABLE V Input breakdown current test - Continued

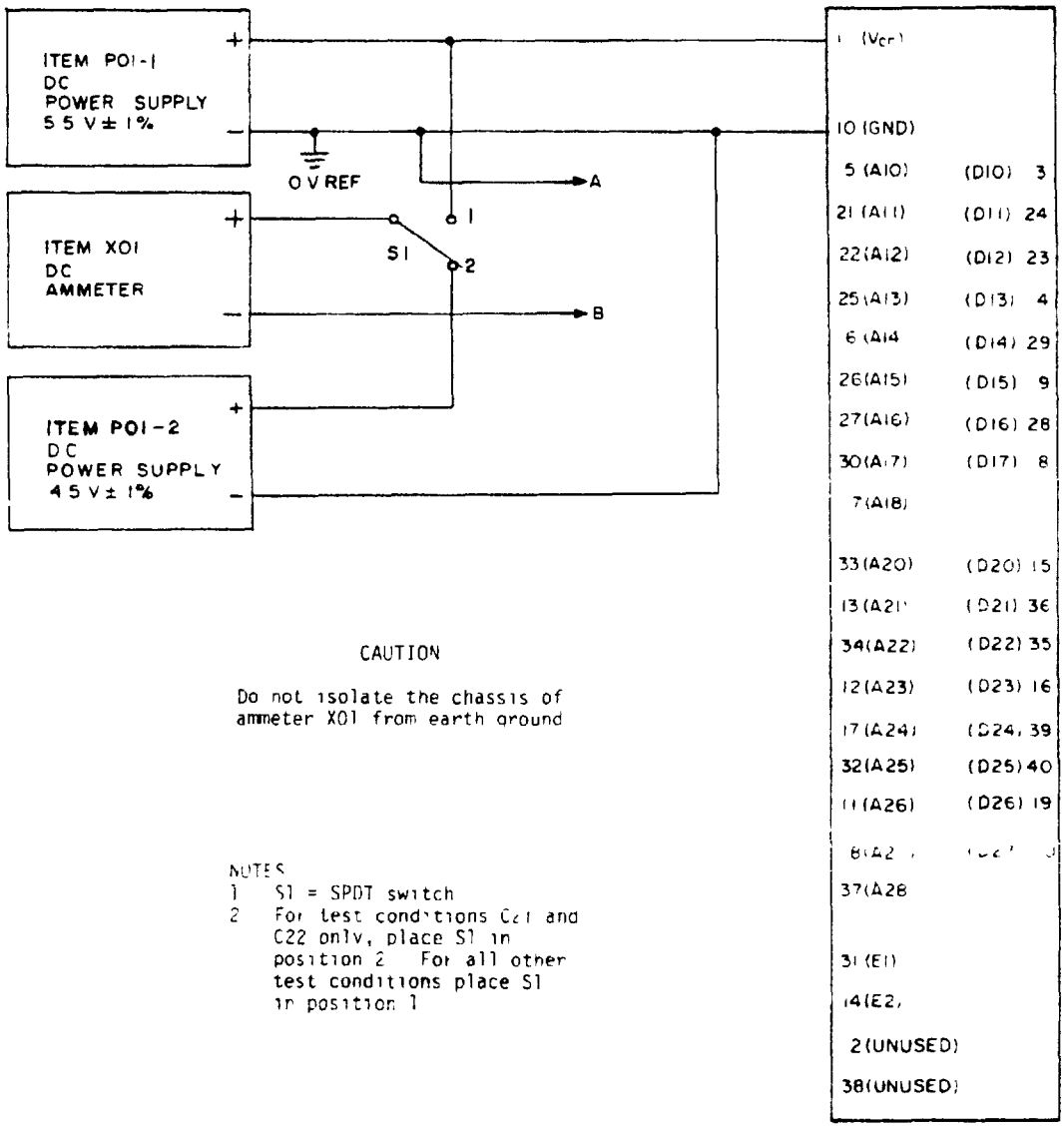
Test condition	Parameter		Input		Initial limit 25° +5°C Ambient		Unit	Test figure
			+ to pin	- to pin	Min	Max		
C8	A17	Input breakdown current	5,6,7,11,12, 13,14,17,18,21, 22,25,26,27,31, 32,33,34,37	30	0 0	2.0	mA	13
C9	A18		5,6,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34,37	7				
C10	A20		5,6,7,11,12, 13,14,17,18,21, 22,25,26,27,30, 31,32,34,37	3?				
C11	A21		5,6,7,11,12,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	13				
C12	A22		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,37	34				
C13	A23		5,6,7,11,13,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	12				
C14	A24	Input breakdown current	5,6,7,11,12,13, 14,18,21,22,25, 26,27,30,31,32, 33,34,37	17	0 0	2 0	mA	13

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TABLE V Input breakdown current test - Continued

Test condition	Parameter		Connect		Initial limits 25 +5°C Ambient		Unit	Test figure
			+ to pin	- to pin	Min	Max		
C15	A25	Input breakdown current	5,6,7,11,12, 13,14,17,18,21, 22,25,26,27,30, 31,33,34,37	32	0.0	2.0	mA	13
C16	A26		5,6,7,12,13,14, 17,18,21 22,25, 26,27,30,31,32, 33,34,37	11				
C17	A27		5,6,7,11,12,13, 14,17,21,22,25, 26,27,30,31,32, 33,34,37	18				
C18	A28		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34	37		2.0		
C19	E1		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,32, 33,34,37	31		40.0		
C20	E2	Input breakdown current	5,6,7,11,12,13, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	14		40 0		
C21	E1	Input breakdown current (4,5V)	5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,32 33,34,37	31		2.0		
C22	F2	Input breakdown current (4,5V)	5,6,7,11,12,13 7,18,21,22,25, 6,27,30,31,32 13,34,37	14	0.0	2 0	mA	13

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CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

NOTES
1 S1 = SPDT switch
2 For test conditions C21 and C22 only, place S1 in position 2. For all other test conditions place S1 in position 1

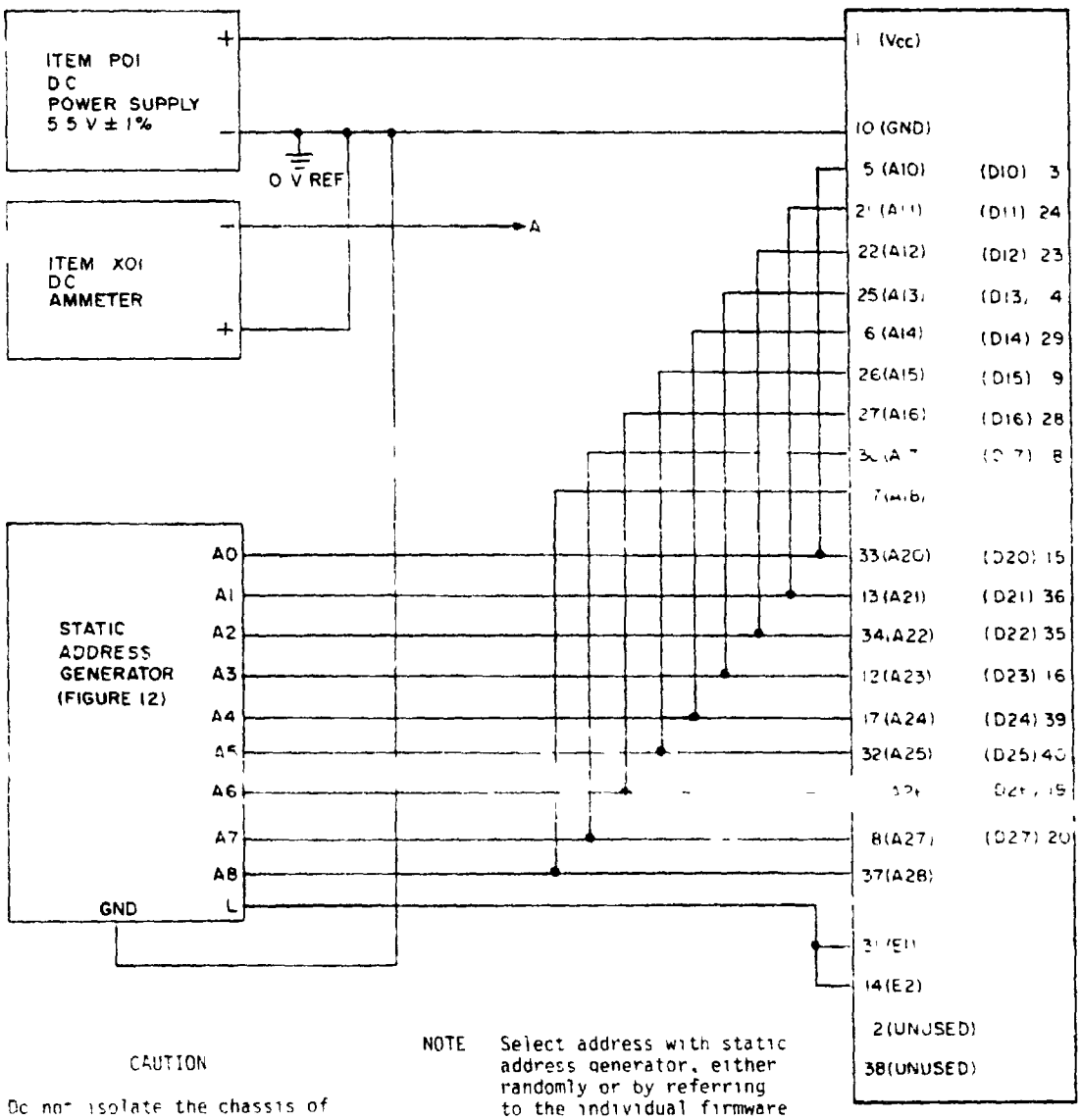
FIGURE 13 Input breakdown current test circuit for table V

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TABLE VI Output short circuit current test.

Test condition	Parameter		Connect to pin	Initial limits 25° ±5°C Ambient		Unit	Test figure
				Min	Max		
D1	D10	Output short circuit current	3	-8.0	-90.0	mA	14
D2	D11		24				
D3	D12		23				
D4	D13		4				
D5	D14		29				
D6	D15		9				
D7	D16		28				
D8	D17		8				
D9	D20		15				
D10	D21		36				
D11	D22		35				
D12	D23		16				
D13	D24		39				
D14	D25		40				
D15	D26		19				
D16	D27	Output short circuit current	20	-8.0	-90.0	mA	14

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CAUTION

Do not isolate the chassis of ammeter X01 from earth ground

NOTE Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a high level. If an output is low for all addresses, omit this test for that output.

FIGURE 11. Output short circuit current test circuit for Table V.

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TABLE VII. Output leakage current (low-level) test

Test conditions	Terminal	Parameter	Device type	Limits 25°C Ambient		Unit	Line figure
				L	Max		
E1	D10	Output leakage current, low level	1	-100	100	µA	15
E2	D11		2				
E3	D12		21				
E4	D1		4				
E5	D14		19				
E6	D15		9				
E7	D16		18				
E8	D17		8				
E9	D20		15				
E10	D21		36				
E11	D22		35				
E12	D23		16				
E13	D24		39				
E14	D25		40				
E15	D26		19				
E16	127	Output leakage current, low level	20	-100	100	µA	15

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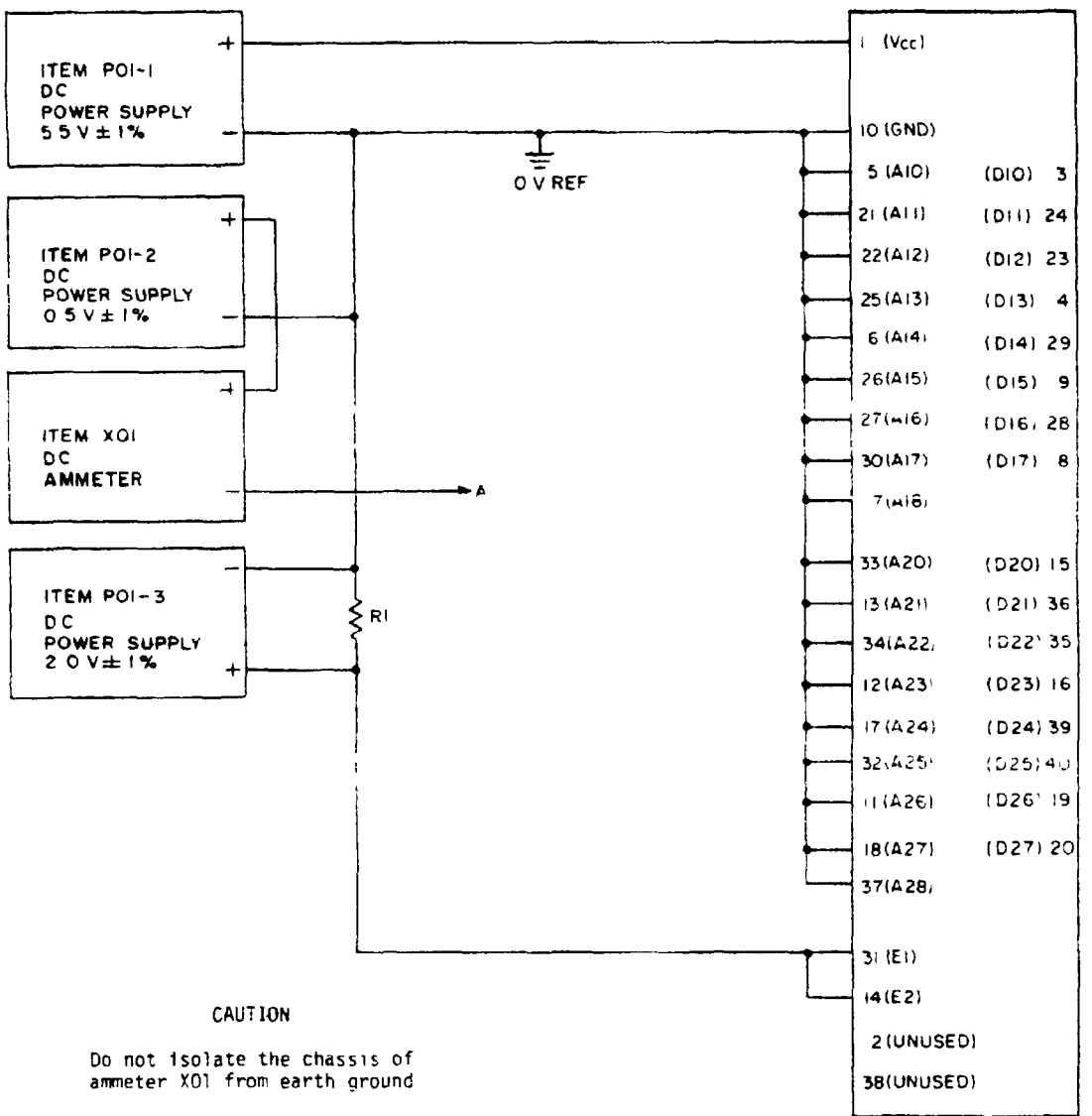


FIGURE 15 Output leakage current (low-level) test circuit for table VI

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TABLE VIII. Output leakage current (high-level) test

Test condition	Parameter		Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
			pin to pin	Min	Max		
F1	D10	Output leakage current, high level	3	-100	100	µA	16
F2	D11		24				
F3	D12		23				
F4	D13		4				
F5	D14		29				
F6	D15		9				
F7	D16		28				
F8	D17		8				
F9	D20		15				
F10	D21		36				
F11	D22		35				
F12	D23		16				
F13	D24		39				
F14	D25		40				
F15	D26		19				
F16	D27	Output leakage current, high level	20	-100	100	µA	16

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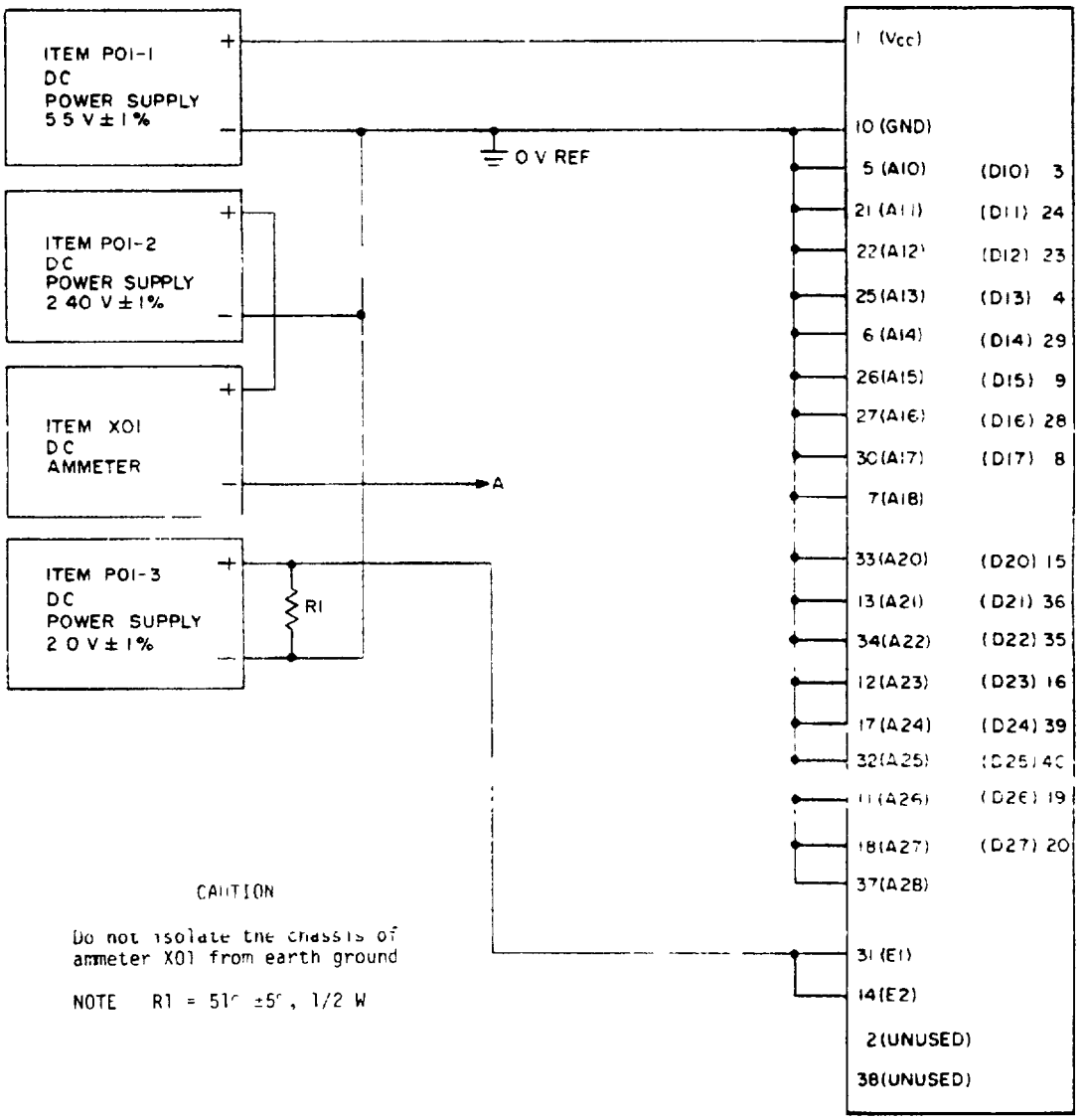


FIGURE 16 Output leakage current (high-level) test circuit for table VIII

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TABLE IV Input clamp diode test

Test condition	Parameter		Connect to pin	Initial limits 25° ±5°C Ambient		Unit	Test figure
				Min	Max		
G1	A10	Input clamp diode	5	-0.2	-1.5	V	17
G2	A11		21				
G3	A12		22				
G4	A13		25				
G5	A14		6				
G6	A15		26				
G7	A16		27				
G8	A17		30				
G9	A18		7				
G10	A20		33				
G11	A21		13				
G12	A22		34				
G13	A23		12				
G14	A24		17				
G15	A25		32				
G16	A26		11				
G17	A27		18				
G18	A28	Input clamp diode	37	-0.2	-1.5	V	17

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TABLE IX Input clamp diode test - Continued

Test condition	Parameter		Connect A to pin	Initial limit 25° ±5°C Ambient Min	Initial limit 25° ±5°C Ambient Max	Unit	Test figure
C19	E1	Input clamp diode	31	-0.2	-1.5	V	17
C20	E2	Input clamp diode	14	-0.2	-1.5	V	17

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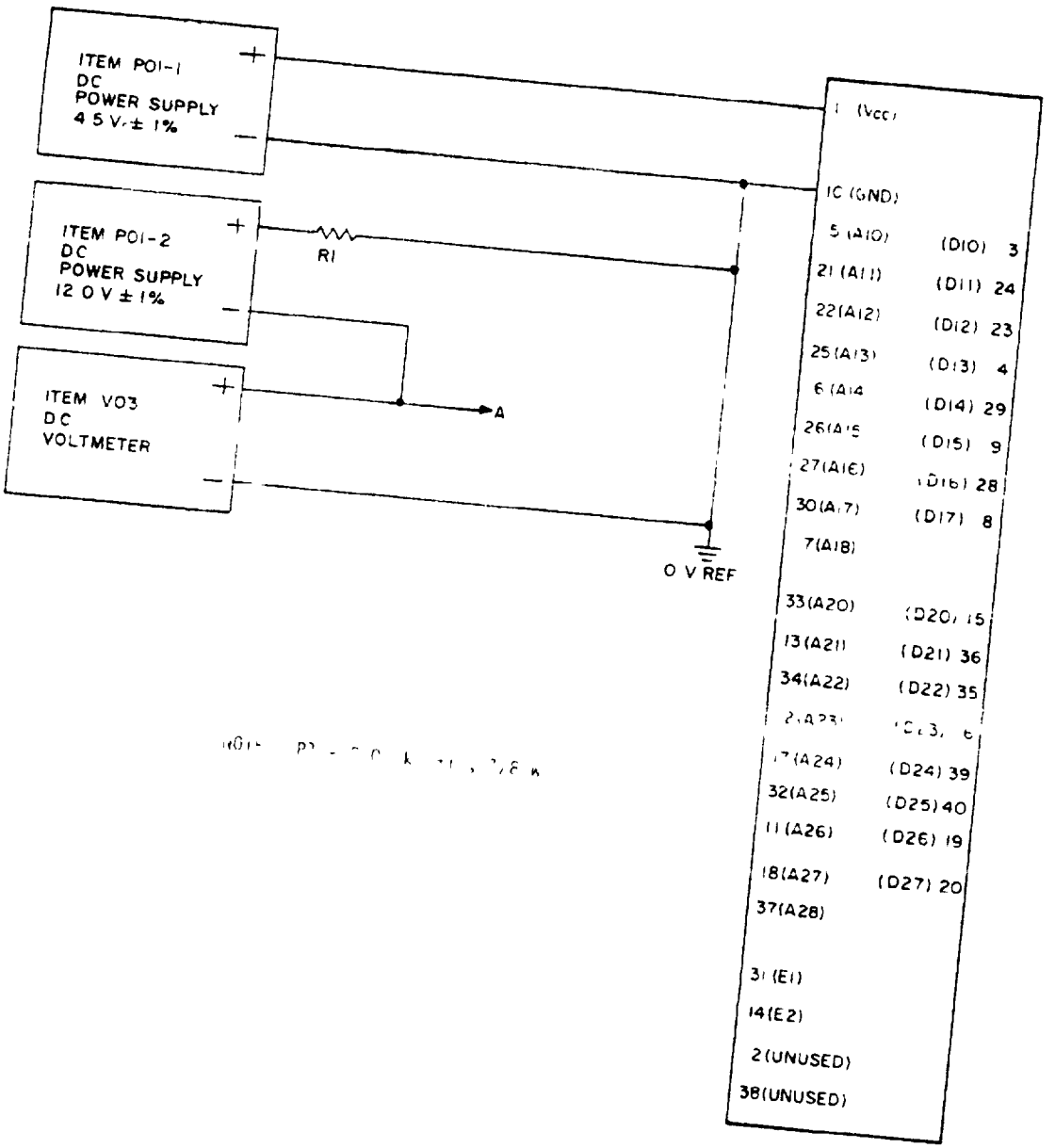


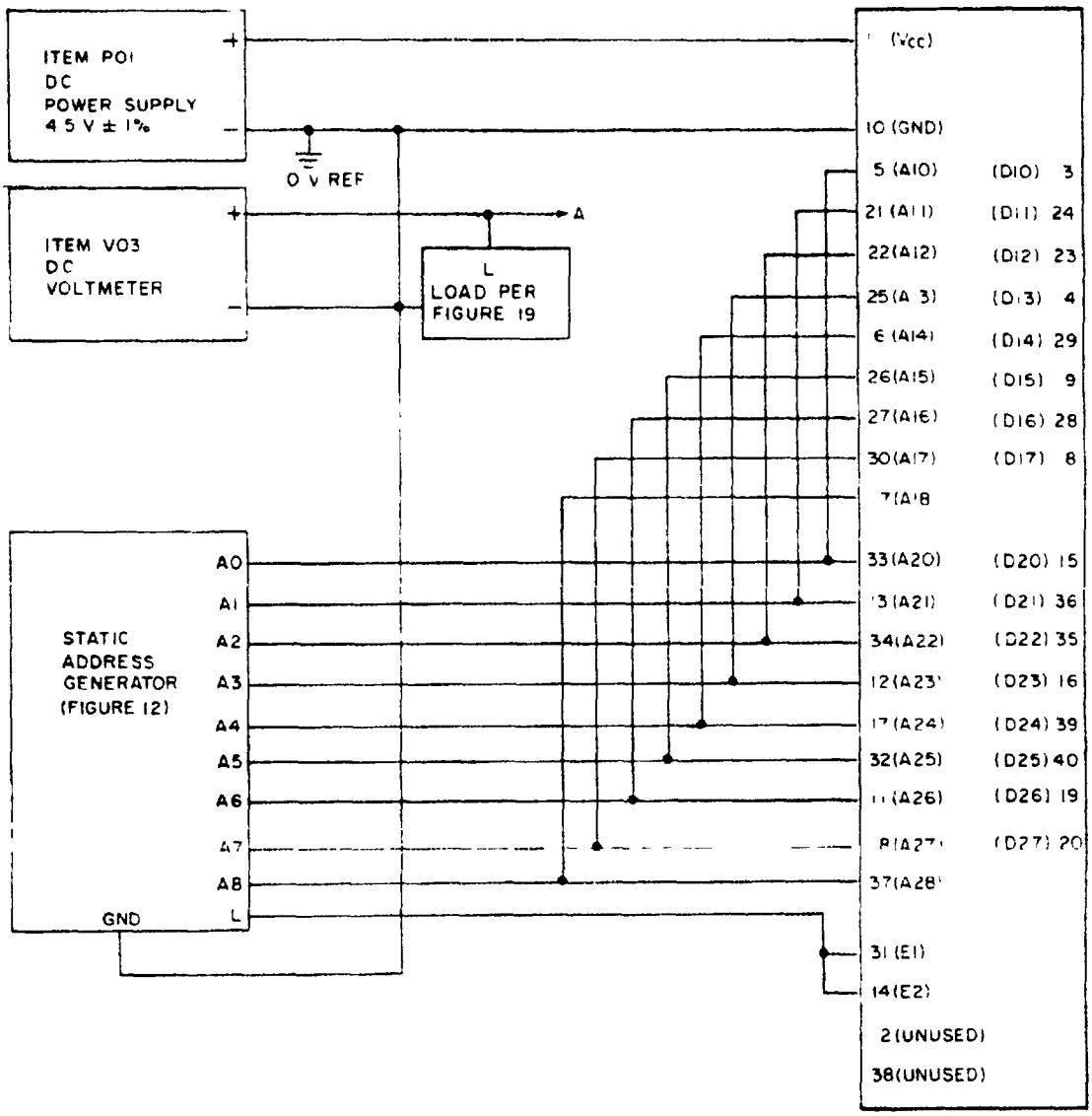
FIGURE 17 Input clamp diode test circuit for table IX

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TABLE X Static output voltage (low-level) test

Test condition	Parameter		Connect pin	Initial limits 5° ±5°C Ambient Min	Max	Unit	Test figure
H1	D10	Static output voltage, low level	3	0.0	0.50	V	18
H2	D11		24				
H3	D12		23				
H4	D13		4				
H5	D14		9				
H6	D15		9				
H7	D16		28				
H8	D17		8				
H9	D20		15				
H10	D21		36				
H11	D22		35				
H12	D23		16				
H13	D24		39				
H14	D25		40				
H15	D26		19				
H16	D27	Static output voltage, low level	20	0.0	0.50	V	18

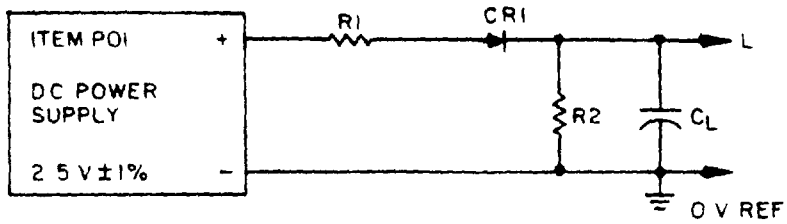
Source: https://assist.dla.mil -- Downloaded: 20260-20T15:54Z
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NOTE Select address with static address generator, either randomly or by referring to the individual firmware pattern such that the output under test is a low level. If an output is high for all addresses, omit this test for that output.

FIGURE 18 Static output voltage (low-level) test circuit for table I

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NOTES
1 CR1 = 1N3600, 1N4150, or equivalent
2 R1 = Select value for a current flow of 8 0 mA ±5 with point L held at 0 5 V
3 R2 = 2 61 k ±1/8 W
4 For test conditions G1-G2
C_L = 50 pF (including probe and fixture capacitance)
5 For test conditions N1-N16, P1-P16, S-R2, S-S2
C_L = 50 ±5 pF (including probe and fixture capacitance)
6 For test conditions N17-N32, P17-P32, P3-R4
C_L = 150 ±5 pF (including probe and fixture capacitance)
7 Capacitor C_L is not required for dc electrical test conditions J1-J16 and for functional test condition T
8 Lead length from load circuit to module output pin under test is to be less than 1 inch (25 4 mm)

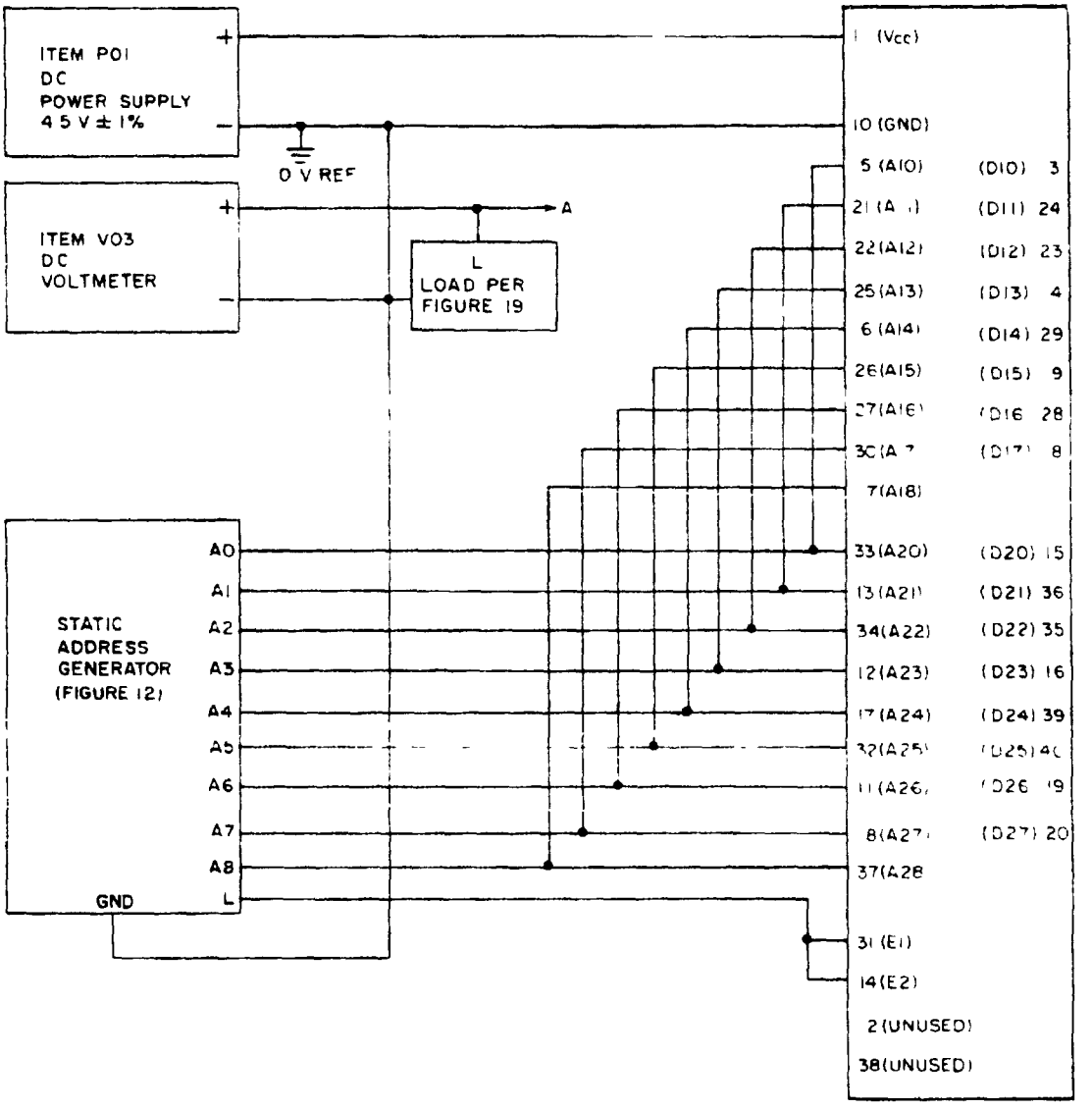
FIGURE 19 Logic load circuit

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TABLE XI. Static output voltage (high-level) test

Test condition	Parameter		Connect V to pin	Initial limit 25° ±5°C Ambient Min	Initial limit 25° ±5°C Ambient Max	Unit	Test figure
J1	D10	Static output voltage, high level	3	2.4	4.5	V	20
J2	D11		24				
J3	D12		23				
J4	D13		4				
J5	D14		29				
J6	D15		9				
J7	D16		28				
J8	D17		8				
J9	D20		15				
J10	D21		36				
J11	D22		35				
J12	D23		16				
J13	D24		39				
J14	D25		40				
J15	D26		19				
J16	D27	Static output voltage, high level	20	2.4	4.5	V	20

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NOTE Select address with static address generator, either randomly or by referring to the individual firmware pattern, such that the output under test is a high level. If an output is low for all addresses, omit this test for that output.

FIGURE 20. Static output voltage (high-level) test circuit for table V.

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TABLE XII Input current (high-level) te...

Test condition	Terminals		Connect		Initial limits -55° +5°C Ambient		Unit	Test figure
			F to pin	B to pin	Min	Max		
K1	A10	Input current, High level	6,7,11,1..,13, 14,17,18 21,22, 25,26,27 30,31, 32,33,34 37	5	-100	80	µA	21
K2	A11		5,6,7,11,12,13, 14,17,18,22,25, 26,27,30,31,32, 33,34,37	21				
K3	A12		5,6,7,11,12,13, 14,17,18,21,25, 26,27,30,31,32, 33,34,37	22				
K4	A13		5,6,7,11,12,13, 14,17,18,21,22, 26,27,30,31,32, 33,34,37	25				
K5	A14		5,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34,37	6				
K6	A15		5,6,7,11,12,13, 14,17,18,21,22, 25,27,30,31,32, 33,34,37	26				
K7	A16	Input current, high level	5,6,7,1 ,12,13, 14,17,18,21,22, 25,26,30,31,32, 33,34,37	27	-100	80	µA	21

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TABLE XII Input current (high-level) test - Continued

Test condition	Test		Terminal		Limits at T_A = +25°C Ambient		Unit	Test figure
			Measured	Supply	Min	Max		
K8	A17	Input current high level	5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,31,32, 33,34,37	30	-100	80	µA	21
K9	A18		5,6,11,12,13, 14,17,18,21,24, 25,26,27,30,31, 23,33,34,37	7				
K10	A20		5,6,7,11,12,13, 14,17,18,21,24, 25,26,27,30,31, 32,34,37	33				
K11	A21		5,6,7,11,12,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	13				
K12	A22		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,37	34				
K13	A23		5,6,7,11,13,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	12				
K14	A24	Input current, high level	5,6,7,11,12,13, 14,18,21,22,25, 26,27,30,31,32, 33,34,37	17	-100	80	µA	21

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:54Z
Check the source to verify that this is the current version before use.

TABLE XII Input current (high-level) test - Continued

Test condition	Test		Input		Initial limits 25 +5°C Ambient		Unit	Test figure
			All pins	1 to pin	Min	Max		
K15	A25	Input current, high level	5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 33,34,37	32	-100	80	μA	21
K16	A26		5,6,7,12,13,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	11				
K17	A27		5,6,7,11,12,13, 14,17,21,22,25, 26,27,30,31,32, 33,34,37	18				
K18	A28		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34	37				
K19	F1		5,6,7,11,12,13, 14,17,18, 1, [illegible], 25,26,27,30,32, 33,34,37	31				
K20	E2	Input current, high level	5,6,7,11,12,13, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	14	-100	80	μA	21

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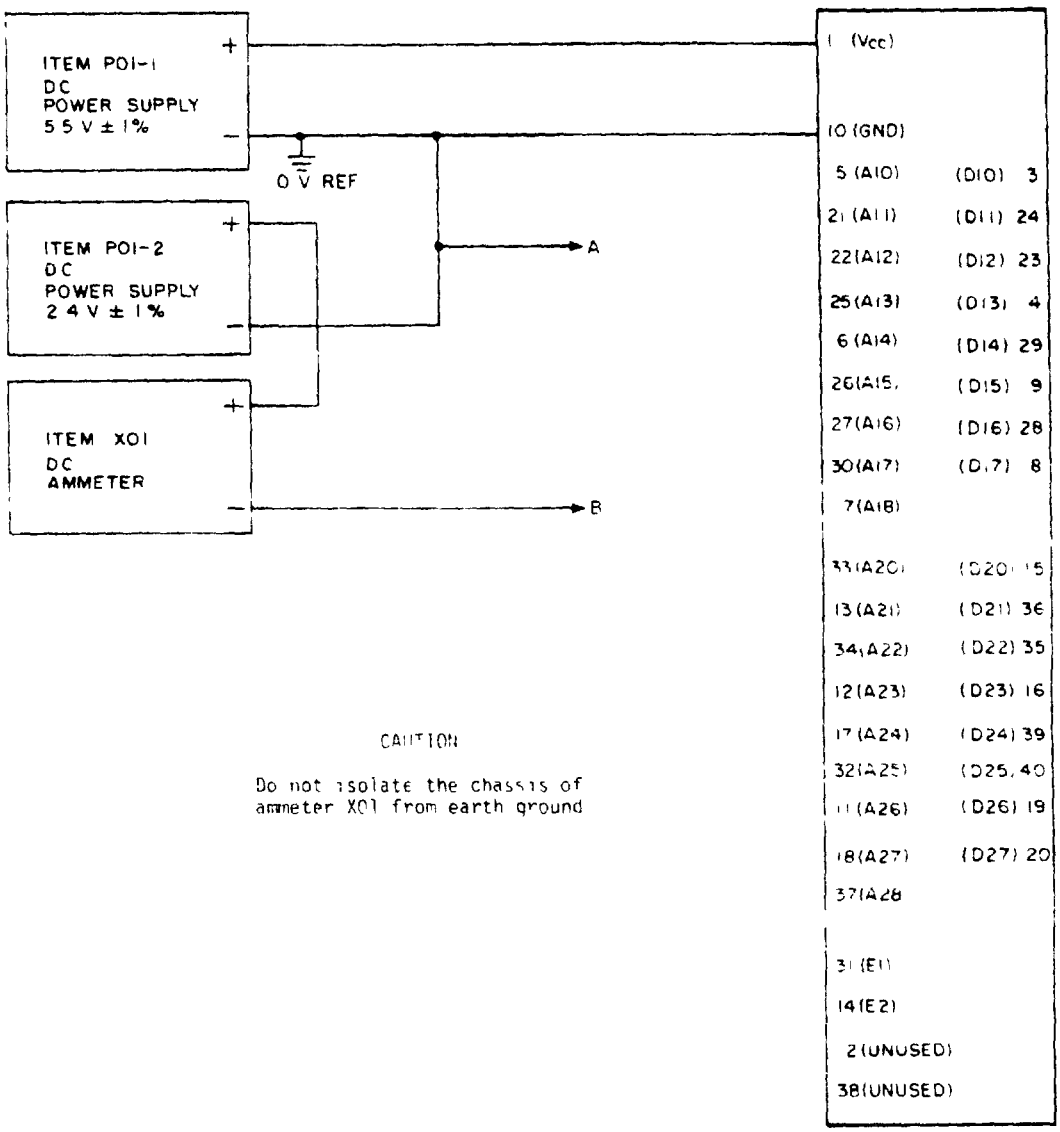


FIGURE 21 Input current (high-level) test circuit for table XII

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TABLE XIII Input current (low-level) test

Test condition	Test		Terminal A (or pin)	Terminal B (or pin)	Initial limit 25° +5°C Ambient Min	Initial limit 25° +5°C Ambient Max	Unit	Test figure
L1	A10	Input current, low level	6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34 37	5	0.0	-0 88	mA	22
L2	A11		5,6,7,11,12,13, 14,17,18 22,25, 26,27,30,31,32, 33,34,37	21				
L3	A12		5,6,7,11,12,13, 14,17,18,21,25, 26,27,30,31,32, 33,34,37	22				
L4	A13		5,6,7,11,12,13, 14,17,18,21,22, 26,27,30,31,32, 33, 4,37	25				
L5	A14		5,7 11,12,13, 14, 7,18,21,22, 25, 6,27,30,31, 32, 3,34,37	6				
L6	A15		5,6,7,11,12,13, 14,17,18,21,22, 25,27,30,31,32, 33,34,37	26				
L7	A16	Input current, low level	5,6,7,11,12,13, 14,17,18,21,22, 25,26,30,31,32, 33,34,3	27	0 0	-0.88	mA	22

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TABLE XIII Input current (low-level) test - Continued

Test condition	Symbol	Measured terminal	Common		Limits of Input 25° 45°C Ambient		Unit	Not to exceed
			a to pin	b to pin	Min	Max		
L8	A17	Input current, low level	5,6,7,11,12,14, 17,18,21,23,25, 26 27,31,32,33, 34,37	30	0 0	-0 88	mA	22
L9	A18		5,6,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33,34,37	7				
L10	A20		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,34,37	33				
L11	A21		5,6,7,11,12,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	13				
L12	A22		5,6,7,11,12,13, 14,17,18,21,22, 25,26,27,30,31, 32,33 37	34				
L13	A23		5,6,7,11,13,14, 17,18,21,22,25, 26,27,30,31,32, 33,34,37	12				
L14	A24	Input current, low level	5,6,7,11,12,13, 14,18,21,22,25, 26,27,30,31,32, 33 34,37	17	0 0	-0 88	mA	22

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TABLE XIII. Input current (low-level) - Continued

Test condition	Parameter		Connect A to pin	Connect B to pin	Initial limits 25° +5°C Ambient Min	Initial limits 25° +5°C Ambient Max	Unit	Test figure
L15	A25	Input current, low level	5,6,7,11,12,13,14,17,18,21,22,25,26,27,30,31,33,34,37	32	0.0	-0.88	mA	22
L16	A26		5,6,7,12,13,14,17,18,21,22,25,26,27,30,31,32,33,34,37	11				
L17	A27		5,6,7,11,12,13,14,17,21,22,25,26,27,30,31,32,33,34,37	18				
L18	A28		5,6,7,11,12,13,14,17,18,21,22,25,26,27,30,31,32,33,34	37				
L19	E1		5,6,7,11,12,13,14,17,18,21,22,25,26 27,30,32,33,34,37	31				
L20	E2	Input current, low level	5,6,7,11 12,13,17,18,21,22,25,26,27,30,31,32,33,34,37	14	0.0	-0 88	mA	22

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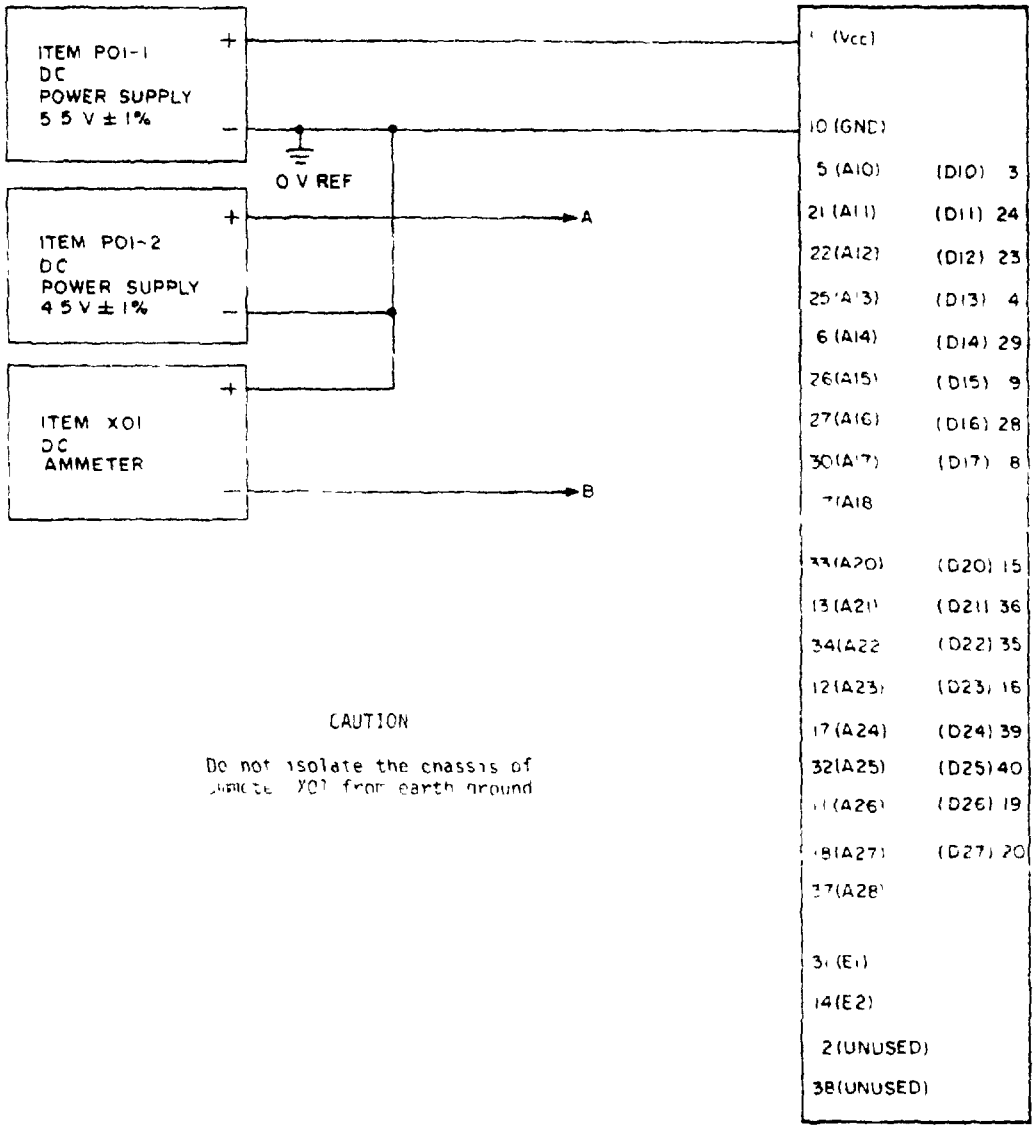


FIGURE 22 Input current (low-level) test circuit for table XIII

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TABLE XIV. Power supply current test

Test condition	Parameter	Initial limits 25°C ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current	5.0	520.0	mA	23

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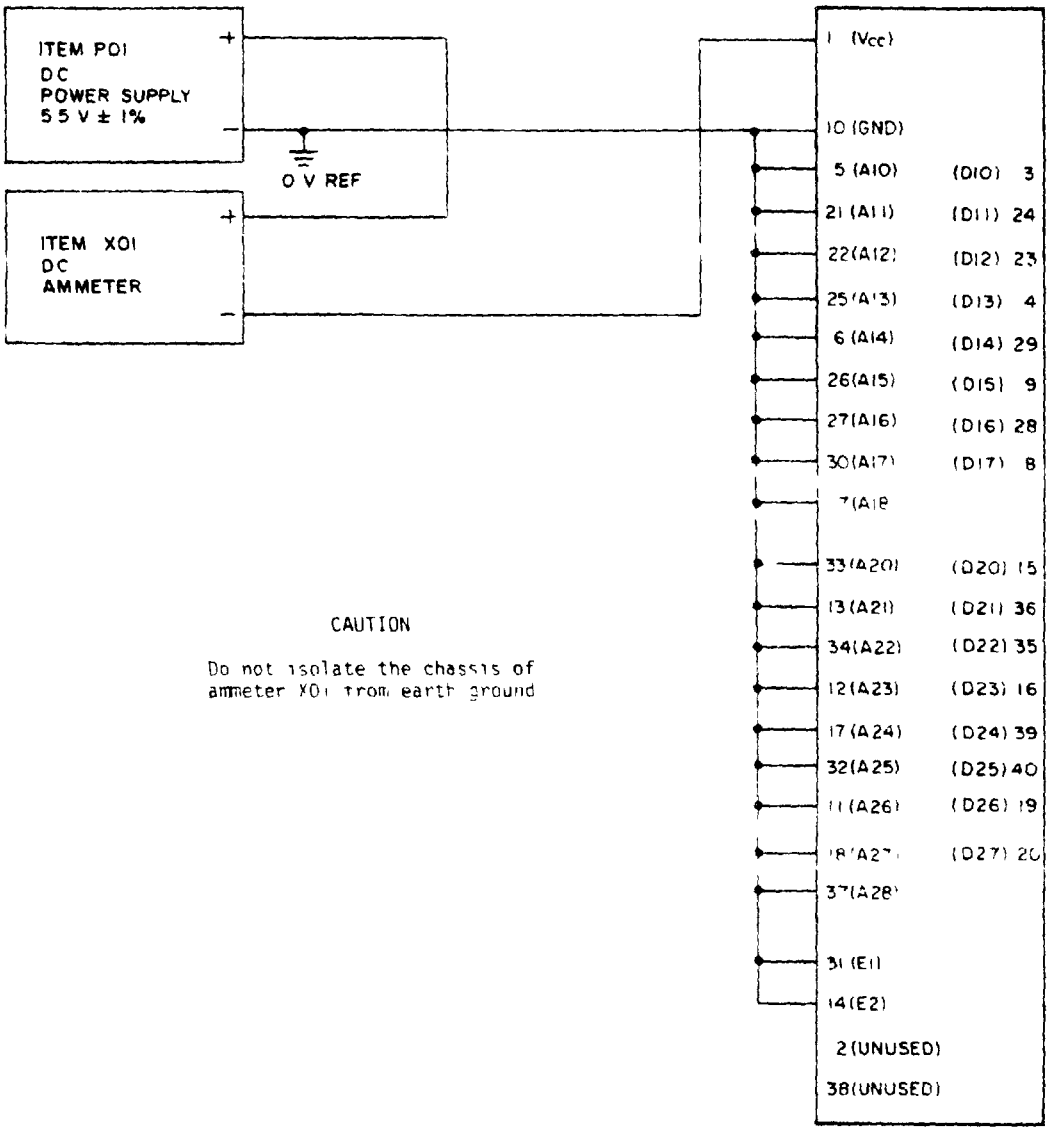


FIGURE 23 Power supply current test circuit for table XIV

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TABLE XV Transition time tests

Test condition	Parameter	Connect		Limits -55°C ≤ T_A ≤ +125°C Ambient		Unit	Test figure
		A to pin	P to pin	Min	Max		
N1	D10 Transition time (t_{TLH}), C_L = 50 pF	See notes 2 and 3, figure 25	3	4.0	40.0	ns	24, 25
N2	D11		24				
N3	D12		23				
N4	D13		4				
N5	D14		29				
N6	D15		9				
N7	D16		28				
N8	D17		8				
N9	D20		15				
N10	D21		36				
N11	D22		35				
N12	D23		16				
N13	D24		39				
N14	D25		40				
N15	D26		19				
N16	D27 Transition time (t_{THL}), C_L = 50 pF	See notes 2 and 3, figure 25	20	4.0	40.0	ns	24, 25

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TABLE IV Transition time tests - Continued

Test condition	Device	Device under test		Device limits -55°C to +125°C Ambient		Unit	Test figure
		Vcc pin	Ground pin	Min	Max		
N17	D10	Transition time (t_{TLH}), C_L = 150 pF	See notes 2 and 3, figure 25	3	4.0	50.0	ns	24, 25
N18	D11			24				
N19	D12			23				
N20	D13			4				
N21	D14			29				
N22	D15			9				
N23	D16			28				
N24	D17			8				
N25	D20			15				
N26	D21			36				
N27	D22			35				
N28	D23			16				
N29	D24			39				
N30	D25			40				
N31	D26			19				
N32	D27	Transition time (t_{TLH}), C_L = 150 pF	See notes 2 and 3, figure 25	20	4.0	50.0	ns	24,25

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TABLE XV Transition time tests - Continued

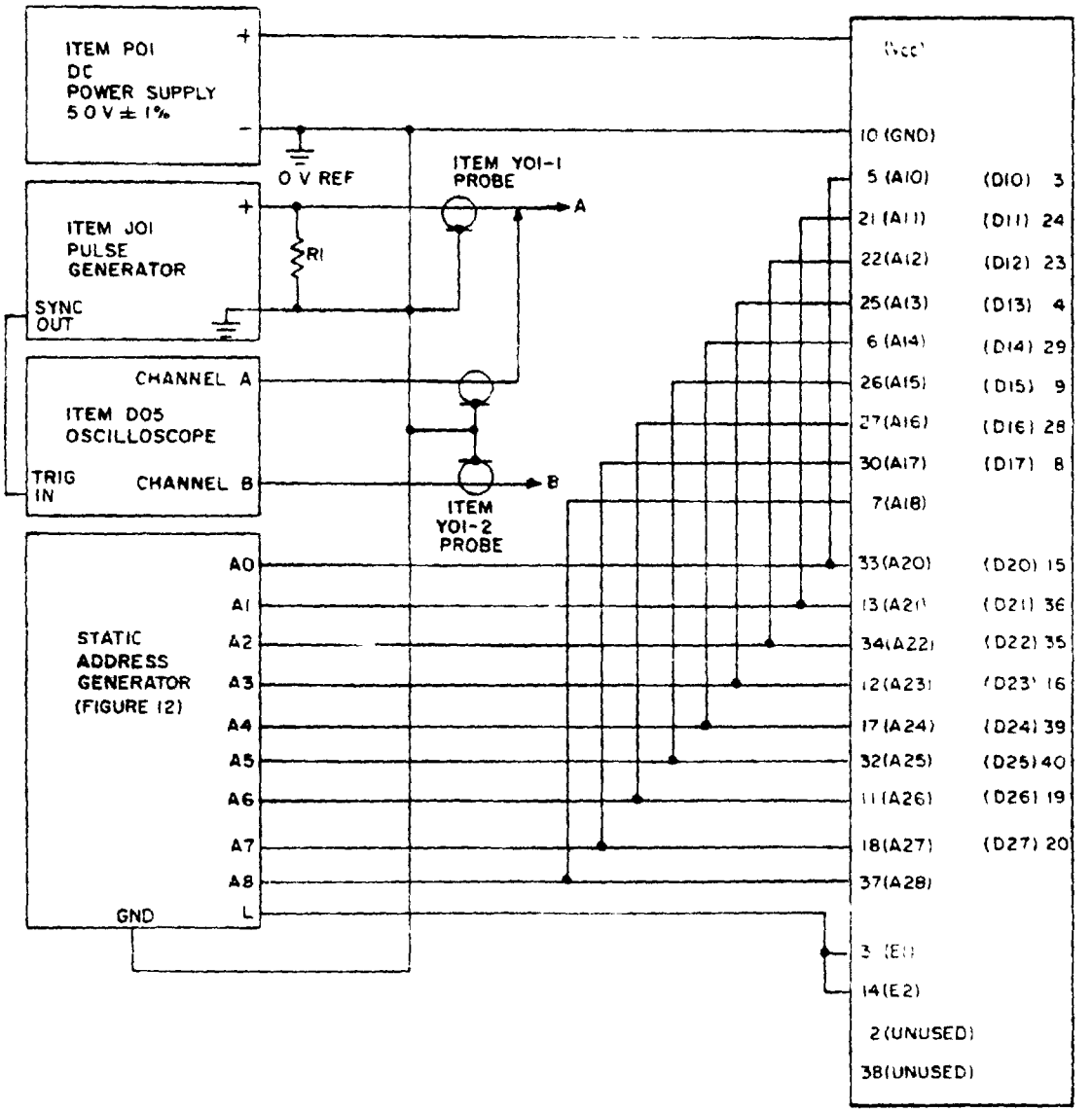
Test condition	Parameter		Connect		Initial limit 25°C ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	D10	Transition time (t_{THI}), C_L = 50 pF	See notes 2 and 3, figure 25	3	4.0	18.0	ns	24, 25
P2	D11			24				
P3	D12			23				
P4	D13			4				
P5	D14			29				
P6	D15			9				
P7	D16			28				
P8	D17			8				
P9	D20			15				
P10	D21			36				
P11	D22			35				
P12	D23			16				
P13	D24			39				
P14	D25			40				
P15	D26			19				
P16	D27	Transition time (t_{THI}), C_L = 50 pF	See notes 2 and 3, figure 25	20	4.0	18.0	ns	24, 25

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TABLE XV Transition time tests - Continued.

Test condition	Test number	Test	Connect A to pin	Connect B to pin	Initial limits 25°C +5°C Ambient Min	Initial limits 25°C +5°C Ambient Max	Unit	Test figure
P17	D10	Transition time (t_{THL}), C_l = 150 pF	See notes 2 and 3, figure 25	3	4.0	20.0	ns	24, 25
P18	D11			24				
P19	D12			23				
P20	D13			4				
P21	D14			29				
P22	D15			9				
P23	D16			28				
P24	D17			8				
P25	D20			15				
P26	D21			36				
P27	D22			35				
P28	D23			16				
P29	D24			39				
P30	D25			40				
P31	D26			19				
P32	D27	Transition time (t_{THL}), C_l = 150 pF	See notes 2 and 3, figure 25	20	4.0	20.0	ns	24, 25

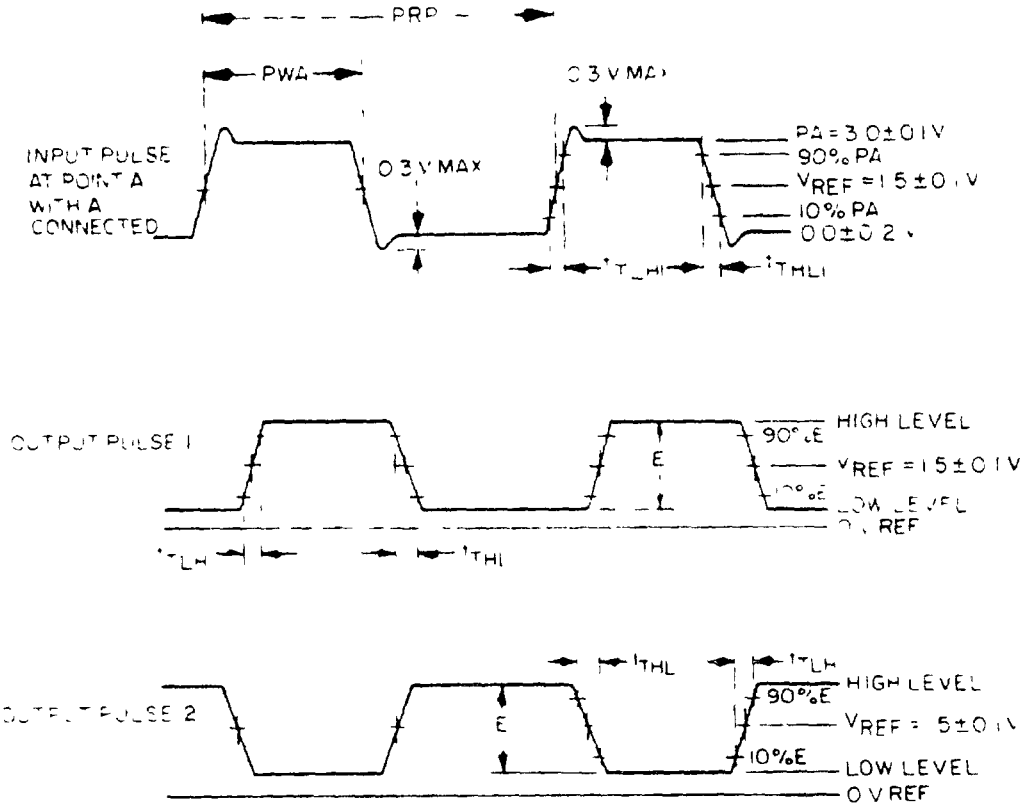
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:54Z
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NOTES
1. R1 = 51Ω ±5%, 1/2 W
2. Load output under test with load as specified on figure 19
3. For test conditions N and P, set the high and low level voltages of the static address generator to
   H = 3.5 -0.5 V, L = 0.0 V

FIGURE 24. Transition time test circuit for table XV

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NOTES

1. Input pulse parameters
   t_{TLH1}, t_{THL1} = 4 ±2 ns
   PWA = 200 ±10 ns
   PRP = 500 ±50 ns
2. Select an address with the static address generator and one address input which will allow (when point A is connected) the output to change from a low level to a high level (output pulse 1 waveform) or a high level to a low level (output pulse 2 waveform). If an output has the same output level for all addresses, omit this test for that output.
3. When point A is connected to an address pin, the corresponding static address generator pin is disconnected.
4. Before making the measurements specified in table XV, verify that the output pulse high level is ≥ 2.4 V and the low level is ≤ 0.5 V.
5. The lead length of electrical conductors between terminator resistor R1 and input pin, and between probes YO1-1, YO1-2 and input and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 2. Transition time waveforms for table XV.

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TABLE XVI Enable recovery time test.

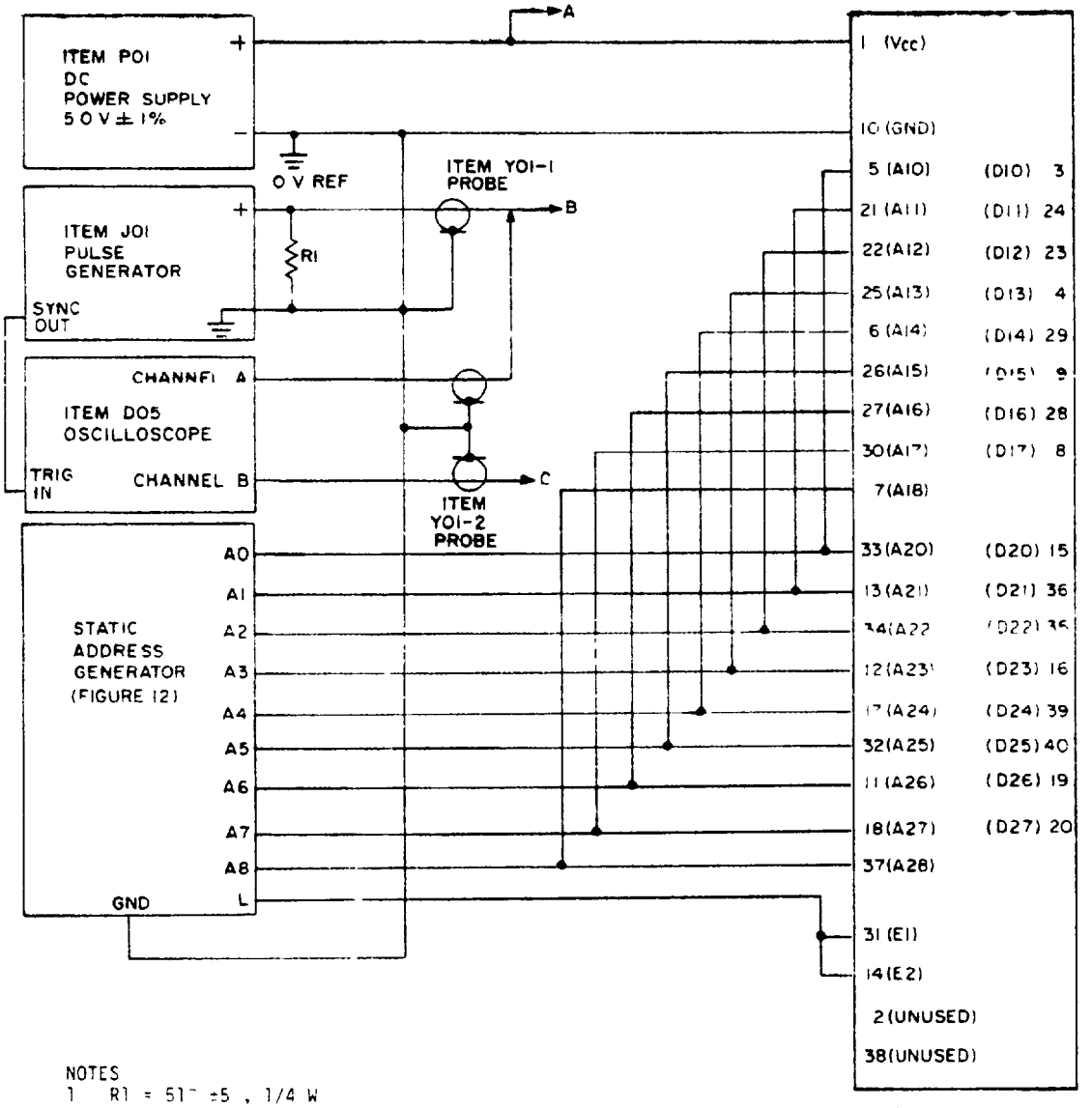
Test condition	Parameter		Connect (See notes 2 and 3, figure 26)			Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q1	D10	Enable recovery time (t_{ERL})	14	31	3	5	50	ns	26, 27
Q2	D11				24				
Q3	D12				23				
Q4	D13				4				
Q5	D14				29				
Q6	D15				9				
Q7	D16				28				
Q8	D17		14	31	8				
Q9	D20		31	14	15				
Q10	D21				36				
Q11	D22				35				
Q12	D23				16				
Q13	D24				39				
Q14	D25				40				
Q15	D26				19				
Q16	D27	Enable recovery time (t_{ERL})	31	14	20	5	50	ns	26, 27

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TABLE XVI Enable recovery time test - Continued

Test condition	Parameter	Connect (See notes 2 and 3, figure 26)			Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q17	D10 Enable recovery time (t_{ERH})	14	31	3	5	50	ns	26, 27
Q18	D11			24				
Q19	D12			23				
Q20	D13			4				
Q21	D14			29				
Q22	D15			9				
Q23	D16			28				
Q24	D17	14	31	8				
Q25	D20	21	14	15				
Q26	D21			36				
Q27	D22			35				
Q28	D23			16				
Q29	D24			39				
Q30	D25			40				
Q31	D26			19				
Q32	D27 Enable recovery time (t_{ERH})	31	14	20	5	50	ns	26, 27

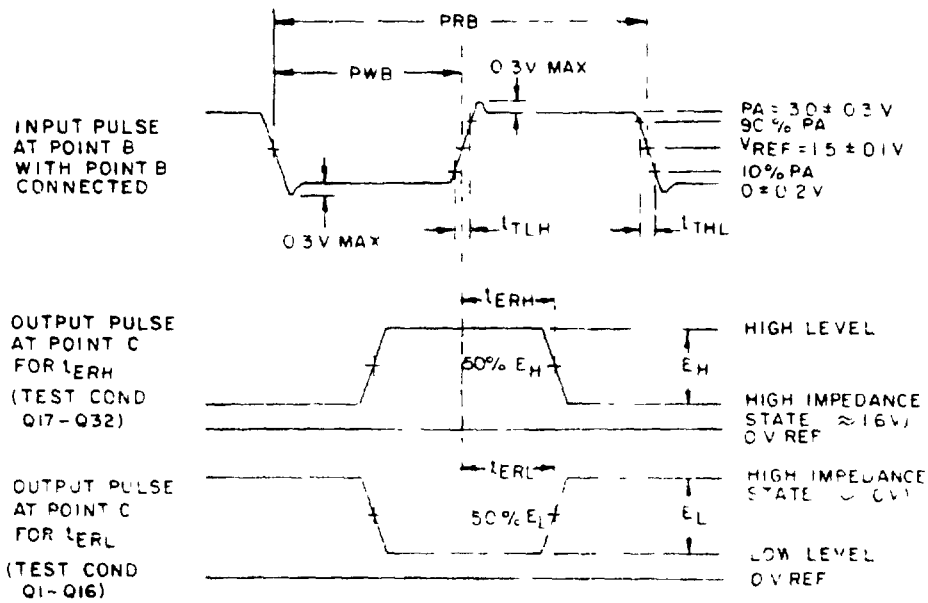
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NOTES
1 R1 = 51 ±5 , 1/4 W
2 For test condition Q only, set the high and low levels of the static address generator to
H = 3.5 ±0.5 V, L = 0 V

FIGURE 26 Enable recovery time tests test circuit for table XVI

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NOTES

1. Input pulse parameter
   PWB = 500 ±50 ns
   PRB = 1,000 ±100 ns
   t_{TLH}, t_{THL} = 10 ±2 ns
2. Select an address with the static address generator and one address input which will allow (when point B is connected) the output to change from a low level to the high-impedance level (t_{ERL}) or a high level to the high-impedance level (t_{ERH}). If an output has a high level for all addresses, omit the t_{ERL} test for that output. If an output has a low level for all addresses, omit the t_{ERH} test for that output.
3. When point B is connected to an enable pin, the static address generator output (L) is disconnected.
4. Load each output with the load as specified in figure 19.
5. Before making the measurements specified in table XVI verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
6. The lead length of electrical conductors between terminator resistor R1 and input pin, and between probes Y01-1, Y01-2 and input pins shall be less than 1 inch (25.4 mm).
7. All voltages are referenced to ground.

FIGURE 27. Enable recovery time waveforms for table XVI.

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TABLE VII. Dynamic test

Test condition	Parameter	Limits -55°C ≤ T_C ≤ +125°C unless otherwise specified		Unit	Test figure
		Min	Max		
R1	Galloping read access test pattern, C_L = 50 pF, V_{CC} = 4.5 V	Module shall conform to figure 29 requirements (see notes 7 and 8)			28,29
R2	Galloping read access test pattern, C_L = 50 pF, V_{CC} = 5.5 V	↓			↓
R3	Galloping read access test pattern, C_L = 150 pF, V_{CC} = 4.5 V	↓			↓
R4	Galloping read access test pattern, C_L = 150 pF, V_{CC} = 5.5 V	Module shall conform to figure 29 requirements (see notes 7 and 8)			28,29

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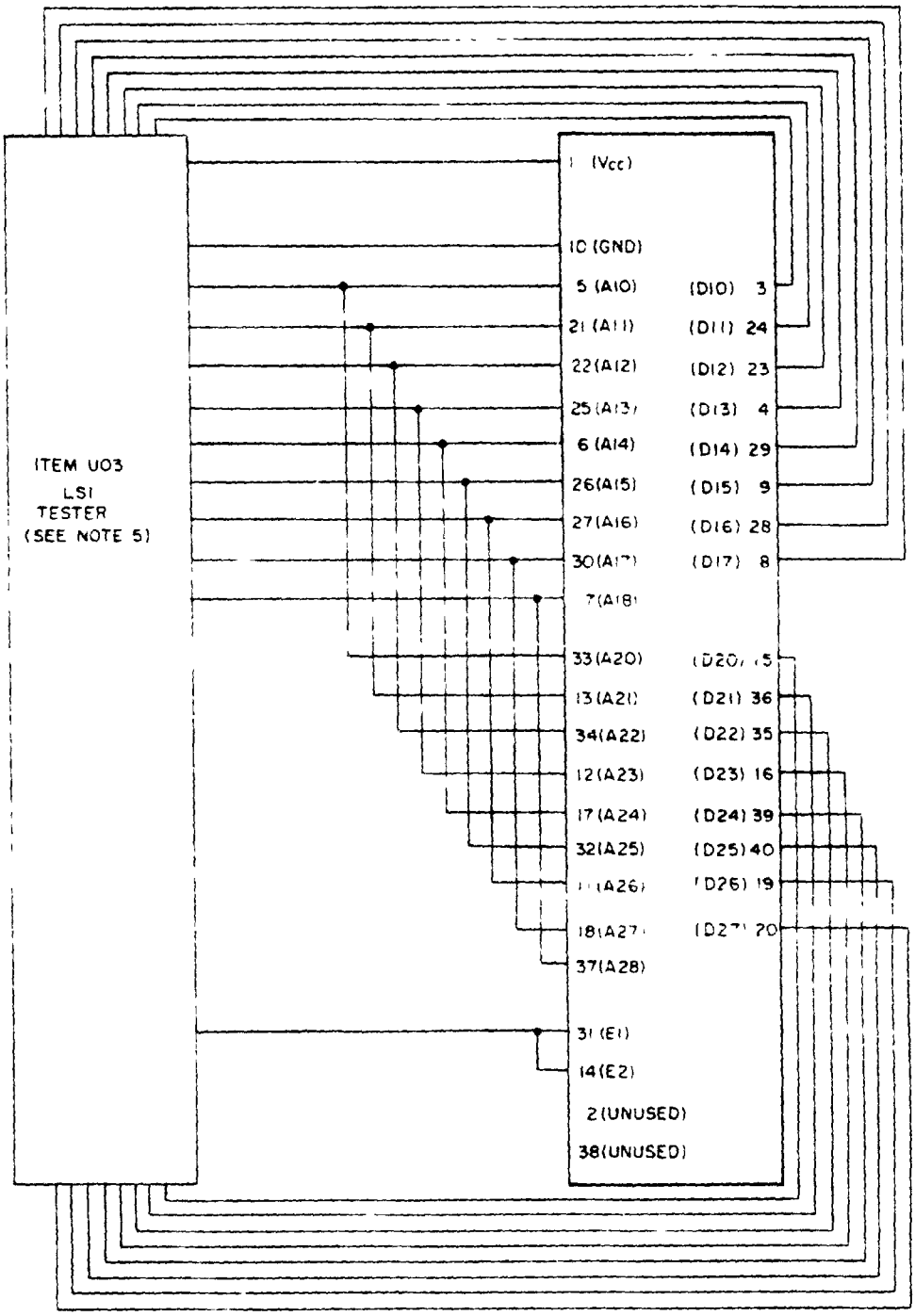


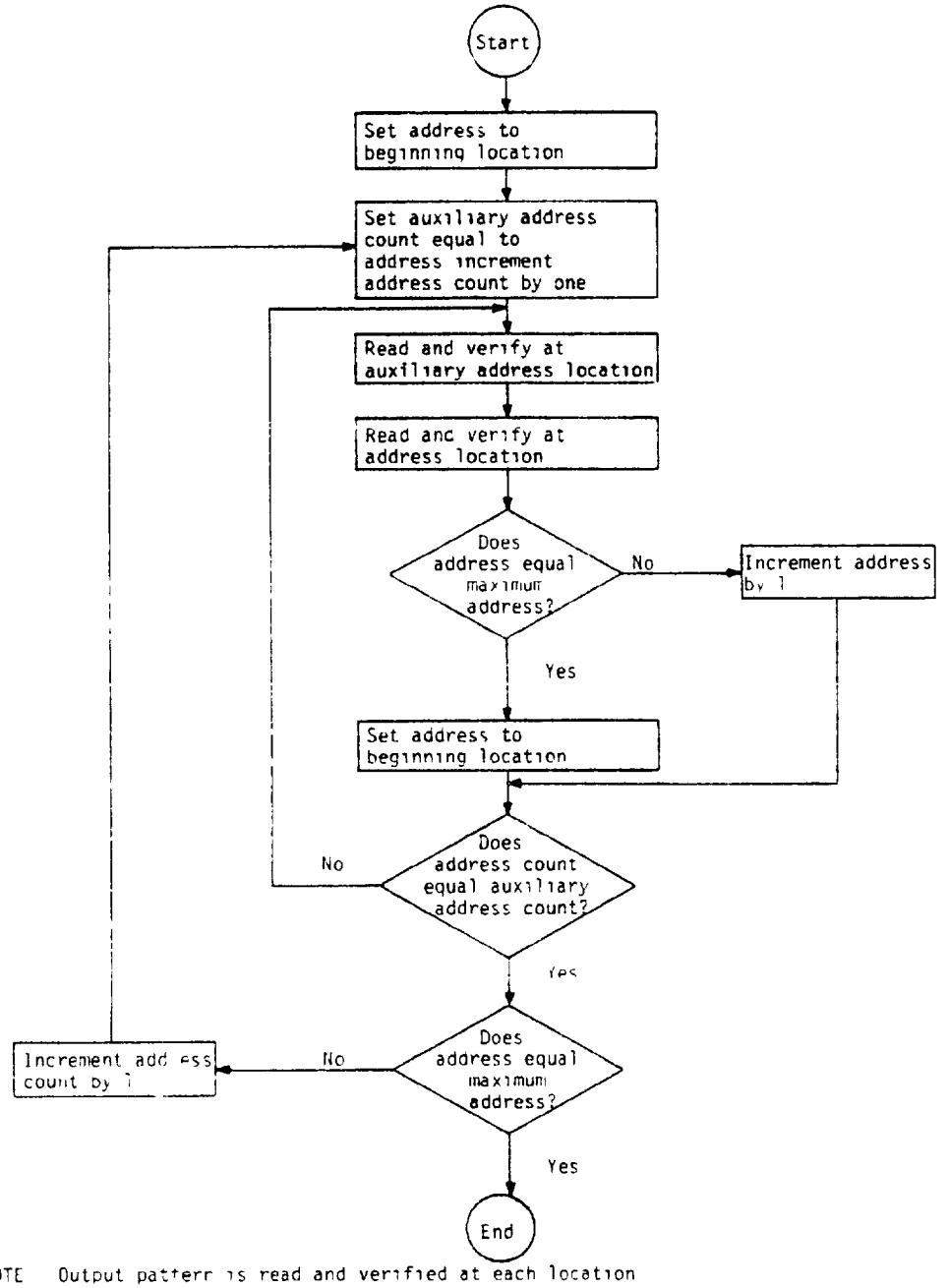
FIGURE 28. Dynamic tests test circuit for table XVII

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NOTES
1. V_{CC} = 4.5 V ±1% for test conditions R1 and R3
2. V_{CC} = 5.5 V ±1% for test conditions R2 and R4
3. Load output under test with load as specified on figure 19 with C_L = 50 pF for test conditions R1-R2
4. Load output under test with load as specified on figure 19 with C_L = 150 pF for test conditions R3-R4
5. See 4.3

FIGURE 2b. Dynamic tests test circuit for table XVII - Continued

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NOTE Output pattern is read and verified at each location

FIGURE 29 Galloping read access pattern program flow chart, test waveforms, and program for table XVII

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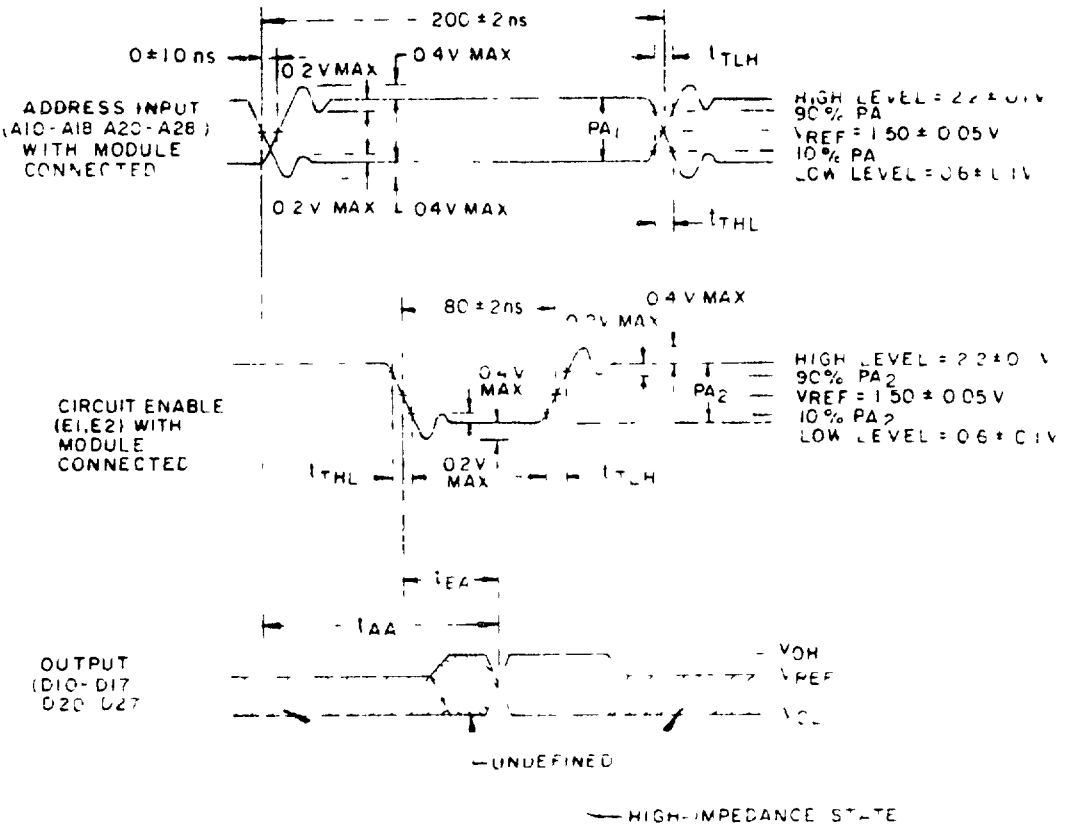


FIGURE 29 Galloping read access pattern program flow chart, test waveforms, and program for table XVII - Continued

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NOTES
1 High-impedance state = 1.6 V
2 Address access time, C_L = 50 pF (t_{AA1}) = 95 +0, -4 ns
3 Address access time, C_L = 150 pF (t_{AA2}) = 105 +0, -4 ns
4 Enable access time, C_L = 50 pF (t_{EA1}) = 45 +0, -2 ns
5 Enable access time, C_L = 150 pF (t_{EA2}) = 55 +0, -2 ns
6 $t_{TLH} = t_{THL}$ = 10 ±4 ns
7 The objective of the galloping read access pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).
8 A program that fulfills the galloping read access test objective would be:

a Read memory address zero
b Read memory address one
c Read memory address zero
d Read memory address two
e Read memory address zero
f Read memory address three
•
•
•
g Read memory address zero
h Read memory address 511
i Read memory address one
j Read memory address two
k Read memory address one
l Read memory address three
•
•
•
m Read memory address one
n Read memory address 511
o Read memory address one
p Read memory address zero
q Read memory address two
r Read memory address three
s Read memory address two
t Read memory address four
•
•
•
u Continue this pattern until all memory addresses have been tested to all other memory addresses

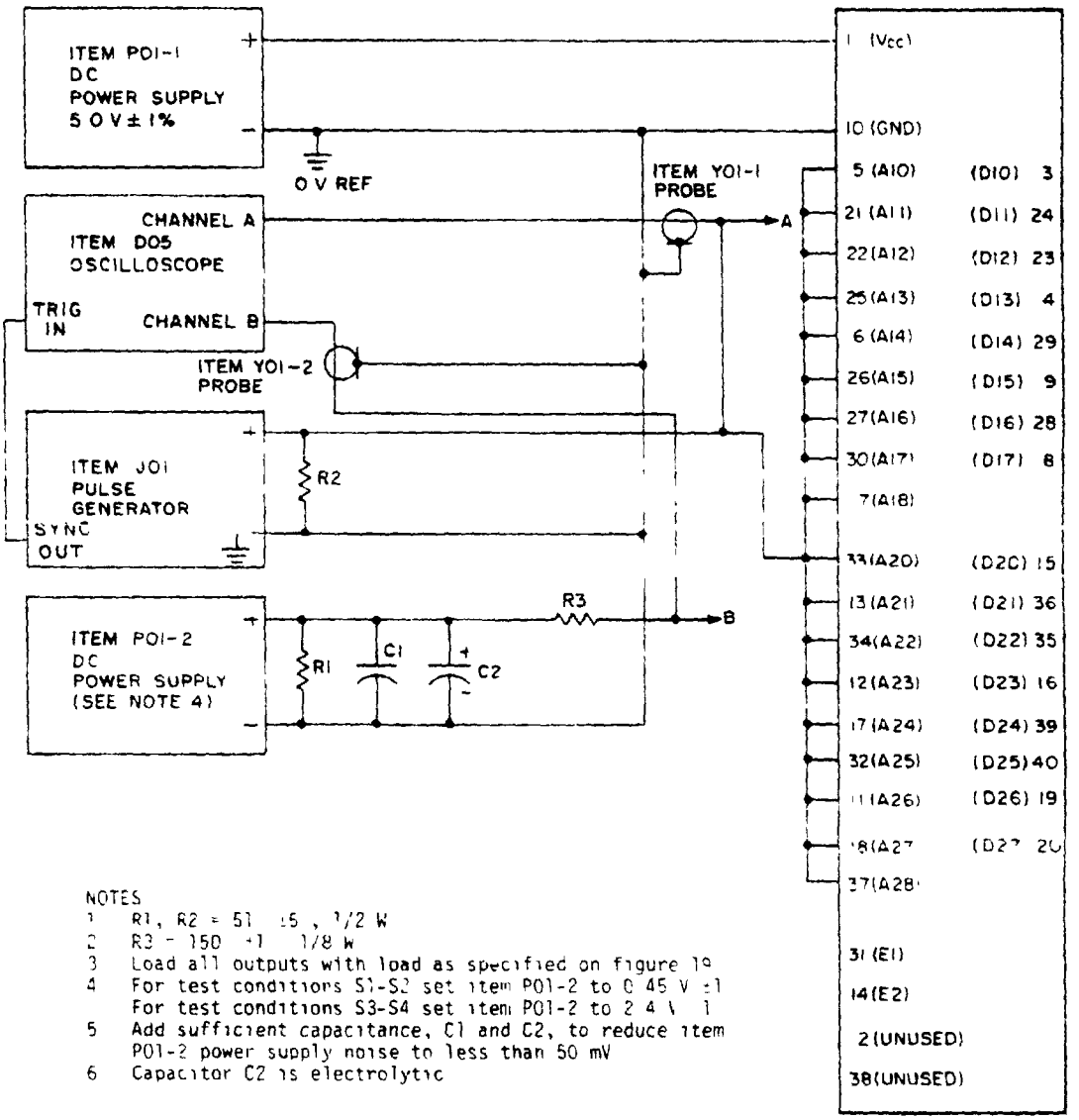
FIGURE 29 Galloping read access pattern program flow chart, test waveforms, and program for table XVII - Continued.

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TABLE XVIII Input crosstalk(low-level and high-level) test

Test condition	Parameter	Connect		Initial limits 25 +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	Input crosstalk, low level	31	14	0	400	mV	30,31
S2	Input crosstalk, low level	14	31	↓	↓	↓	↓
S3	Input crosstalk, high level	31	14	↓	↓	↓	↓
S4	Input crosstalk, high level	14	31	0	400	mV	30,31

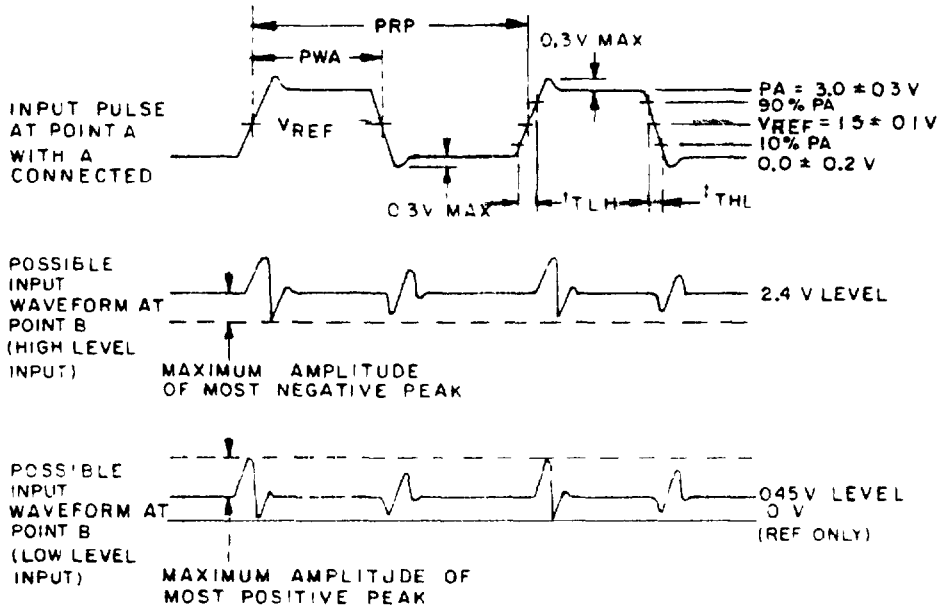
Source: https://assist.dla.mil -- Downloaded: 20260-20T15:54Z
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NOTES
1. R1, R2 = 51 ±5 , 1/2 W
2. R3 = 150 ±1 1/8 W
3. Load all outputs with load as specified on figure 19
4. For test conditions S1-S2 set item P01-2 to 0.45 V ±1
   For test conditions S3-S4 set item P01-2 to 2.4 V ±1
5. Add sufficient capacitance, C1 and C2, to reduce item P01-2 power supply noise to less than 50 mV
6. Capacitor C2 is electrolytic

FIGURE 30 Input crosstalk (low-level and high-level) test circuit for table XVIII

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NOTES
1. Input pulse parameters
   t_{THL}, t_{TLH} = 20 ±2 ns
   PWA = 500 ±50 ns
   PRP = 1,000 ±100 ns
2. The lead length of electrical conductors between terminator resistors and input pins, probes Y01-1, Y01-2 to input pins, load circuits to output pins, and ground leads of probes Y01-1, Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm)

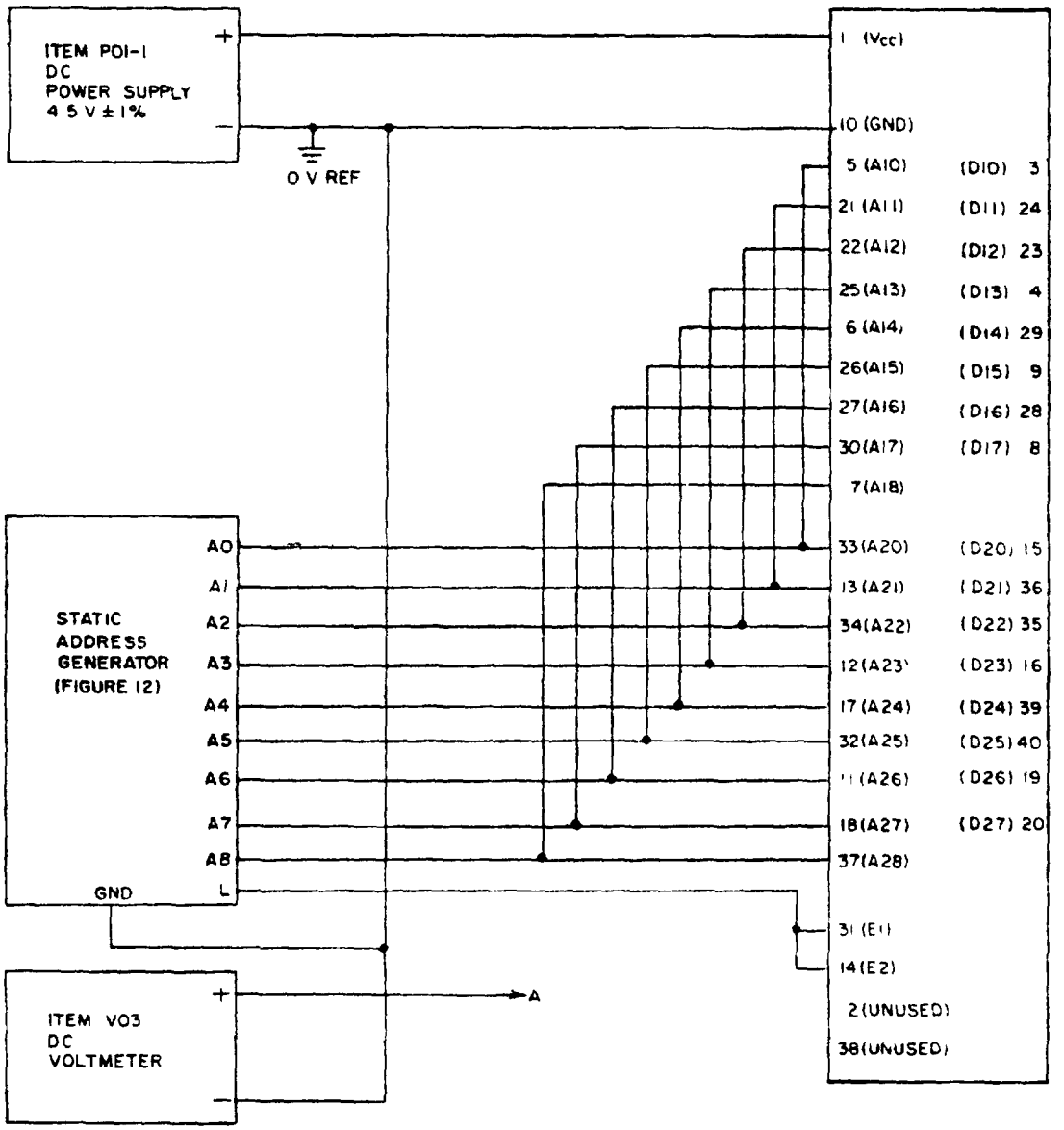
FIGURE 31. Input crosstalk (low-level and high-level) test waveforms for table XVIII

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TABLE XIX Functional test

Test condition	Parameter	Initial limits +5° ±5°C Ambient		Unit	Test figure
		Min	Max		
T	Functional	See note 1, figure 32			32

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NOTES
1 The static address generator is used to select, in turn, each of the 512 address combinations. For each address combination the output is monitored at point A to verify the firmware pattern for the module. Output high level shall be ≥ 2.4 V and low level shall be ≤ 0.5 V.
2 Load the output under test with the load as specified on figure 19.

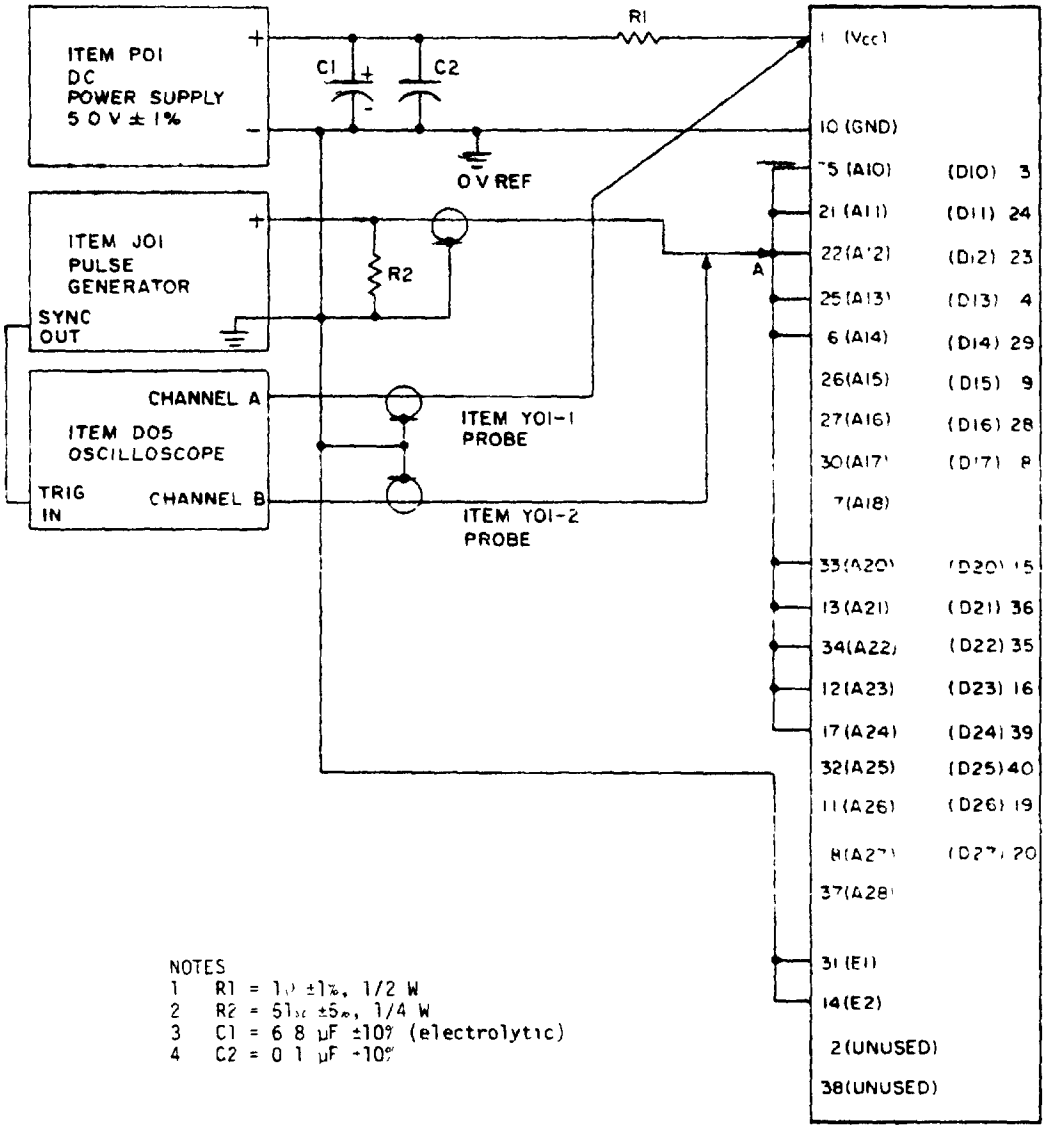
FIGURE 32 Address input threshold voltage (low-level and high-level), test circuit for table XIX

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TABLE XX Power supply transient amplitude test

Test condition	Parameter	Initial limits 25° +5°C Ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	33, 34

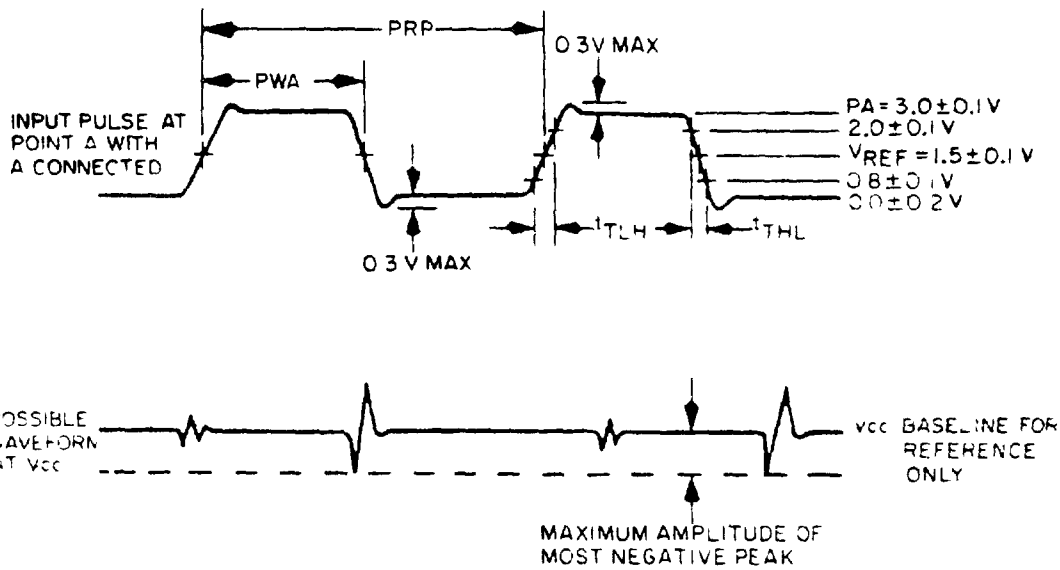
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NOTES
1 R1 = 1.0 ±1%, 1/2 W
2 R2 = 51.0 ±5%, 1/4 W
3 C1 = 6.8 µF ±10% (electrolytic)
4 C2 = 0.1 µF +10%

FIGURE 33 Power supply transient amplitude test circuit for table XX

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NOTES:
1. Input pulse parameters
   t_{TLH}, t_{THL} = 3.5 ±1.0 ns verified at all inputs
   PWA = 1.0 ±0.1 µs
   PRP = 2.0 ±0.1 µs
2. All voltages are referenced to ground
3. Transient amplitude is measured from the steady state V_{cc} level to the most negative excursion of the waveform

FIGURE 34 Power supply transient amplitude waveforms for table XI

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TABLE XXI Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limit 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	16	30	µF	35

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TABLE XXII Input, output capacitance test

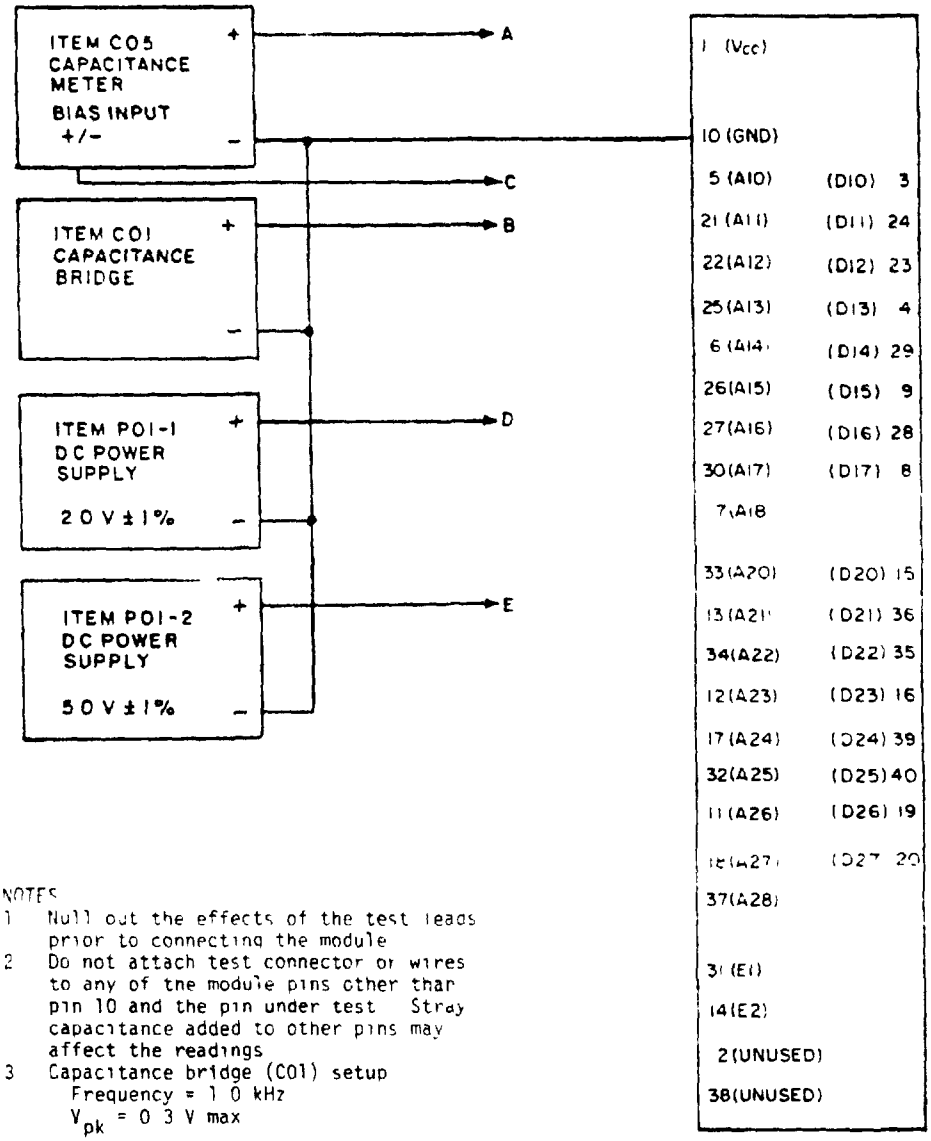
Test condition	Symbol	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° +5°C Ambient Min	Initial limits 25° +5°C Ambient Max	Unit	Test figure
W1	A10	Input capacitance	5	Open	2.0	30.0	pF	35
W2	A11		21					
W3	A12		22					
W4	A13		25					
W5	A14		6					
W6	A15		26					
W7	A16		27					
W8	A17		30					
W9	A18		-					
W10	A20		33					
W11	A21		13					
W12	A22		34					
W13	A23		12					
W14	A24		17					
W15	A25		32					
W16	A26		11					
W17	A27		18					
W18	A28	Input capacitance	37	Open	2.0	30.0	pF	35

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TABLE XXII. Input, output capacitance test - Continued

Test condition	Parameter		Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
W19	E1	Input capacitance	31	Open	2.0	35.0	pF	35
W20	E2	Input capacitance	14			35.0		
X1	D10	Output capacitance	1			20.0		
X2	D11		24					
X3	D12		23					
X4	D13		4					
X5	D14		29					
X6	D15		9					
X7	D16		28					
X8	D17		8					
X9	D20		15					
X10	D21		36					
X11	D22		35					
X12	D23		16					
X13	D24		39					
X14	D25		40					
X15	D26		19					
X16	D27	Output capacitance	20	Open	2.0	20.0	pF	35

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NOTES
1. Null out the effects of the test leads prior to connecting the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings
3. Capacitance bridge (C01) setup
   Frequency = 1.0 kHz
   V_{pk} = 0.3 V max
4. For test conditions X1-X16 only, connect C to D and connect E to module pin 1 (V_{CC})

FIGURE 35 Power supply filter capacitance, and input, output capacitance test circuit for tables XXI and XXII

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